

# Universe24: a local integer law inside the GameBoard, a non-local read-out above it, and what the clicks recover of nature

Alon Gonen

Independent researcher

alon@defounder.ai ORCID 0009-0008-2701-4599

September 2026

## Abstract

Inside a cubic GameBoard one update law of bounded integers propagates records as a beam by algebraic formulas at every Node and interval: one map with a rate and a wall per component, its rates and walls made of six integer operations. Outside, the game above the board, there are detectors and their clicks only, and the beam passes to them with amplitudes at one comparison, the click, the one non-local step and the only read-out; the paper assumes the GameBoard is read in no other way (P11). Physics above the board is arithmetic on counts and the law beneath it is modern algebra, and the paper says why: the rules put into the cells are, written down, a cyclic group, a group ring, integer matrices, an evaluation at the roots of unity and a shift, and every result is an identity of that algebra, exact on the board or in a named limit, compared with the register's readings by kind. Inside, the law's consequences are derived and need no experiment; Outside, the clicks recover known forms and are compared with known experiments, result by result, in one ledger, and the paper shows how far Outside represents reality. Exact on the GameBoard: the conservation of the books and the lattice form of the continuity equation, Gauss's law of a free family's flux under its world condition, the pace of every direction with its Manhattan bound, the click's weight a positive quadratic form of power 2 under one axiom of the apparatus, the exact marginals of a pair, the CHSH sum as an exact function of the grain, 181/64 at the powers of two from 512 to 8192, and Planck's and de Broglie's relations as identities under the declared dictionary. In the limit of the grain: the value  $c = 1/\sqrt{3}$ , Born's rule, Tsirelson's bound, Young's spacing. In the limit under an average that covers the shell, the condition part of the claim: Newton's and Coulomb's inverse square, the retarded potential and Poisson's equation. What no rule forces is said in the same ledger: the Lorentz factor (how the clicks arrive at it is the frame), the bending of light, the values of the masses, the harmonic constants. After a detector the simulator confirms the exact results and the pace; against nature three readings pass, a fourth under the clock's assumed word, twelve fail and one is not compared, the failures the law's own. One prediction:  $S = 181/64$ , inside Poh et al. at 1.05 standard errors.

## 1 Introduction

**The general formula.** The law is one map  $\Phi$  applied at every Node of the GameBoard at every interval, per component  $s$  of the state, each with its own rate  $r$  and wall  $d$ :

$$s \leftarrow s + r, \quad e \leftarrow \text{sign}(s) \min(\lfloor |s|/d \rfloor, a), \quad s \leftarrow s - e d. \quad (1)$$

Each component of the state is a bounded integer accumulator with a declared rate and a declared wall: it is translated by its rate, counted in walls (capped at  $a = 1$  for the drive, uncapped

elsewhere), and reduced by the walls it holds; every wall counted is an event, a Link crossed, a phase step taken, a count completed, a birth, a push, and nothing else happens. The rates and walls are made of six integer operations and nothing else: the translation, the multiplication by a declared integer matrix or bilinear form, the sum in the group ring of the phase circle, a permutation, the evaluation of a record at the roots of unity with its norm, and the division with the remainder kept together with the comparison that is the event. One comparison, the click, is the only read-out, and only a detector's reading is a measurement. Section 2 gives the full definition; every derivation of this paper names which of these rules it starts from.

**Why physics behaves like modern algebra.** The claim of the paper is that the forms of several different physical phenomena are recovered from this one discrete update law and a small number of assumptions, Inside by derivation and Outside at the clicks, and the reason is stated in five steps that the paper then follows. First, Outside is arithmetic in  $\mathbb{Q}$ : every measured quantity is a count at a click or a ratio of two counts (a velocity is Nodes apart over counts apart; a visibility, a correlation and a redshift are ratios of counts), and nothing measured is a real number. Second, Inside is integer and local: six operations on bounded integers at a Node and its six neighbours, each rule generic, vector and local; written down, the rules put into the cells are the cyclic group of the phase, the group ring of the arrivals, the integer matrices of the split and the rotation, the evaluation at the roots of unity and the translation group's shift, so the state is a free  $\mathbb{Z}$ -module and the interval a  $\mathbb{Z}$ -linear map on it (Section 2). Third, the couplings are tables, the circle's  $C$  and  $S$ , the split's and the rotation's integer matrices, the family table, and Born's rule is a coupling computed from  $N_\phi$ , the rungs of Eq. (9), so a click is a comparison of integers. Fourth, the transformations between click families that preserve the passage of information are the Lorentz group up to scale (Section 9), so the symmetry Outside has is the algebra's and not a metric put in. Fifth, every result of the paper is therefore an identity of that algebra, checked against the register's readings by kind and not chosen to fit a reading; where a form was declared, the paper says so. The decisive question is how much of that the law really forces: if every phenomenon needed a rule that puts its result in beforehand, the law would explain nothing; if the same rules force several results with little freedom, that is the contribution. So the paper keeps one ledger (Table 1): for every result, the rules of Eq. (1) it starts from, the assumptions added beside them with their kind, the freedom left, and the ground on which the derivation stands. What can be established now is an explicit model, exact and conditional results and checks after a detector; the broad claim that Outside represents nature still needs the completions Section 9 names, and the paper is built around the distance between the two.

**How the simulator led to the results.** The simulator ran first and the derivations followed it. Every run was registered with its expectation written before the run, its world file and its source fingerprint, and each registered number was then traced back to the operations of the map until a derivation stood or the number was marked a finding. Where a derivation closed, the paper says "derived from the update rules implemented in the simulator"; a relation found in runs alone is a numerical finding; a relation defined in advance is a definition or an input.

**The map of results.** Proved: the theorems of Sections 3 to 7 (the group of the six Ports, the pace bound, the lattice Gleason form, the isometry and the injectivity of the interval, the exact marginals, the finite- $N_\phi$  Bell value). Shown numerically after a detector: the pace and its isotropy, the click's cells,  $S$  at seven grains, Malus's law, the clock's field at two distances (Section 8). Open, with the check that decides each: the closing of Newton's inverse square after a detector (the

equivalence measured, series D3) and Poisson's law, the clock of a body in flight, the bending of light, the values of the masses, the harmonic constants (Section 9).

**The reading rule and the notation.** Only a detector's reading is a measurement (the frame's assumption, P11): a click, a record's moments or an external thing's reading, as the world file declares. Every physics name in this paper (Gauss's, Newton's, Coulomb's, Born's, Planck's, Tsirelson's) names the known form that the law's result recovers, Inside by derivation or Outside at a detector. A number of the GameBoard's state is a diagnostic of the host, never compared with nature and never pinned. Every symbol is named in English at its first use, a scalar plain, a vector in bold lowercase, a matrix or an operator in bold uppercase; the symbol table is Appendix B, the program's glossary is on the tree [12], and every number has its one source [5].

## 2 The law and its implementation

**The GameBoard as a system of information transfer.** The GameBoard is a system of information transfer, and its only objects are messages, the rows: a row is a tuple (Node, direction, age, phase, emitter, content per unit, record, label, amount, multiplicity), and a Link is a channel that carries at most one step of a row per interval in each direction. Nothing else exists Inside: no field at a Node, no register, no memory beyond the rows present. Each interval does six things to messages and nothing else: it moves a row one Link along its digital line (the translation), turns its phase (the translation on the circle), splits it at a splitter into rows with integer weights (the multiplication by a declared table), rotates a labelled pair (an integer matrix), merges the rows that meet at a Node with equal words, opposite phases cancelling (the addition), and, at a detector, reads a record once by one comparison and deletes it (the evaluation and the count). The books say that no message is lost or made in transit: released equals in transit plus absorbed plus escaped plus cancelled at every interval. Two messages of this paper: a row on the heading  $+x$  crosses its Links at the intervals 1, 3, 5, 7, 8 (the hand-worked update below), nothing kept at a Node; and at a balanced splitter one row of amount 1 becomes two of amount 1 and multiplicity 2, which meet again at a second splitter, at one port with equal phases (the words add) and at the other  $N_\phi/2$  apart (the message cancels), so one detector reads the record, 64 of 64 births, and the dark port is the cancelled message (series L, Section 8). Written down, a message is a basis element of a free  $\mathbb{Z}$ -module, the passage is the translation group's shift, the split and the rotation are integer matrices, the merge is the group ring's addition and the read-out is one bilinear form: the passage from physics to modern algebra is the passage from messages with attributes to vectors with coordinates. The lattice gases [30, 31] carry bits on links with collision tables, the cellular automata machines of [24] and the quantum cellular automata of [57] are local rules on cells, and Shannon's channel [7] is the message and its capacity; the GameBoard is of that family, with two differences: its messages carry a phase on a bounded circle and an integer amount, and its only read-out is the click.

**The road from the cells to the algebra.** How the things of physics entered: the amount  $w$  of a row is a positive integer, translated and merged; the phase  $p$  is an element of  $\mathbb{Z}_{N_\phi}$ , turned per Link, added in the group ring, evaluated at the roots of unity; the multiplicity  $\mathbf{m}$  is a positive integer,  $A = \sum a_i^2$  at a split; the age  $\tau$  is a count at the rate 1; mass is the content  $M$ , a count of units per family and an input of the family table, which sets the drive's wall  $N_l N_w M + |p_a|$  (the division), scales the push's bilinear form and the release rate, and is the rest energy  $E_0 = N_l N_w M$ ; charge is  $\rho$ , the declared charge per unit of content, entering only the coupling matrix  $\rho_A \rho_B$  of the push;

momentum is an integer vector, translated by the push, read by the drive as a rate against a wall, its square compared and never rooted; the field a body reads is the label flow  $\mathbf{a} = \sum \text{amount} \times \mathbf{u}_D$ , bilinear in the arrivals; the family table (content, cost per phase step, charge, the strong column, lifetime, phase rate, hand) is read and never computed; and the couplings, the circle's tables at 1/256, the split and rotation tables and the click's Gram matrix, are integer matrices and one bilinear form.

**The families in the algebra.** Every family of the register is one row of declared integers (P8) and nothing else: the quantum  $h$  (1 on a free family, whose rows carry their energy as turns, 0 on a paid family, whose content is the mass), the charge per unit of content as a rational pair, the columns with their signs, the lifetime, whether the phase turns, the hand; the content  $M$  of a body is the world's declaration. Written down, a family is where its quantities live: the content on  $\mathbb{Z}$ , the non-compact scale, free; the phase on  $\mathbb{Z}_{N_\phi}$ , quantised; the charge and a column's coefficient as entries of the coupling matrix; the hand a pseudoscalar under the 48; a family's rows on the fan under the 48; a pair's arms in  $\mathbb{Z}^2 \otimes \mathbb{Z}^2$  under the settings' integer matrices; and the six verbs are the only operations on them. The mass angle  $m$  of the frame is the rest pair's turn,  $2\pi n/d$  per interval, a world's declaration and not the family's. Table 7 (Appendix D) lists every family of the register with its declared integers, its object, when it entered, the verbs that act on it and the click that reads it; a cell the tree does not name says so.

**Rules, parameters, initial conditions.** The law is stated as rules, apart from the parameters and the initial conditions a world supplies. The rules: (P1) every quantity of the state is a bounded integer and every operation integer arithmetic, no real number, root or floating point at run time; (P2) space is a cubic array of Nodes, each joined to six neighbours by Links through directional Ports, a Node's complete local information its NodeState, time a sequence of intervals; (P3) the rates and walls of the map are made of the six operations of Section 1 and nothing else; (P4) a rule reads its own record and the arrivals at its Node and its six neighbours, already delivered, and nothing kept at a Node beyond the events there, no register, remainder or draw; (P5) no row or body crosses more than one Link per interval; (P6) a record is read out once, by one comparison, at the click, the one step that deletes, and the click of a pair is one gather of one record from both settings, the law's one non-local operation. The parameters (P7, P8): the grains, the circle  $N_\phi$ , the pace's grain  $N_l$ , the sphere of directions, the birth wheel  $N_u$ , the fan's grain, the clock's pair, and the roundings declared at load, among them the circle's tables at 1/256; and the physical inputs, the family table (per family the content, the cost  $h$  per phase step, the charge per unit of content, the strong column, the lifetime, the phase rate and the hand; its instance on the tree [10]) and the world's width  $N_w$ , which carries what physics calls Newton's constant. The initial conditions: the box and its boundaries, the bodies with their momenta and contents, the lamps, the detectors, the splitters and the gates of the world file. Four rules are chosen among few (P9) and are assumptions, not derivations: the flight is blind to the crowd, a row's rate in transit constant and its wall the direction's  $T_D$ ; a body's drive is the flight at the fraction its momentum earns,  $|p_a|/(N_l N_w M + |p_a|)$  per axis; the collision's permutation is the cyclic shift; and a clock counts the age moment of the rows at its Node (the potential's form, the owner's word of record 394 [6]) rather than their presence. One axiom of the apparatus (P10): the click stores one pointer per set, the least-rank member of the family Theorem 4 forces, which is Born's form with the fundamental harmonic; which harmonic is a relabelling of the declared rate, and the harmonic constants are not derived. One assumption of the frame (P11): the GameBoard, Inside, is read only through its emitters and detectors, Outside; every result Outside is a click, a number of the board's state

is never a reading, and every comparison with experiment is a comparison of clicks with readings; a measurement is itself an event of the board, an emitter putting a record on it and a detector ending the record at the click, the one step that deletes Inside and the one row that Outside gains. Nothing outside this list is assumed.

**The map.** Here  $s$  is one accumulator of the state, a bounded integer on a circle or a line;  $r$  its declared rate, signed, a constant of the world for a message in transit and a reading of what arrives for a body;  $d$  its declared wall, a positive integer;  $a$  the component's cap, 1 where a cap is declared and  $a = \infty$  where none is (the whole part; the primitive's `at_most` = 0); and  $e$  the event count: every unit of  $e$  is an event (a Link crossed, a phase step taken, a count completed, a birth, a push), and nothing else happens; a reversed rate first cancels what was accumulated the other way. For a count without cap the third step is the division toward zero with its signed remainder kept, the Euclidean division for  $s \geq 0$ ; under the cap the residual above the wall is kept and fires one event at each following interval until spent. The cap  $a = 1$  is the drive's alone, one Link per interval; every other count takes the whole part and can fire several events in one interval, the turn, the owed count, the release, the lamp, the push, so that a rate of 5 against a wall of 2 fires 0, 2, 5, 7, 10 events over the intervals, as the owner's high-rate runs read (record 340 [6]). The 0/1 form  $e \leftarrow [s \geq d]$ , one event bit per interval, holds exactly where the rate never exceeds the wall: the flight ( $2S_1N_l \leq 2T_D$ , at most one Link per interval), the drive by its cap and with the sign ( $e \in \{-1, 0, 1\}$ ), the birth wheel (the rate 1 against  $N_u$ ) and a clock whose rate is below its wall (the derivation's section 0 [19]). The rates and walls of every component are made of six operations and nothing else [18, 16]:

1. the translation of an accumulator by its rate (the flight, the phase, the age, every count);
2. the multiplication by a declared integer matrix or bilinear form (a split, a rotation; a coupling, the click's Gram matrix);
3. the addition in the group ring  $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$  (the merge, where opposite phases cancel);
4. the permutation of the joint state (a collision, a gate);
5. the evaluation of the record at the  $N_\phi$ th root of unity,  $\mathbb{Z}[\mathbb{Z}_{N_\phi}] \rightarrow \mathbb{Z}[\zeta_{N_\phi}]$  ( $N_\phi$  the phase circle's width), and its norm (the click's one quadratic step);
6. the division toward zero with the remainder kept (the Euclidean division for a non-negative accumulator), and the comparison (the carry that is the event; the cap; the threshold).

Two places are not linear and there is no third: the carry, a bijection  $s \mapsto (e, s - ed)$  with  $|s - ed| < d$  and the remainder's sign that of  $s$ , which loses nothing, and the click's threshold, the one comparison read out per record, whose remainder is not read: the one place the law deletes by design. The cap and the sign are operation 6's comparison,  $\min([|s|/d], 1) = [|s| \geq d]$ ; the capped carry is the injection  $s \mapsto (e, s - ed)$ , which loses nothing either, so there is still no third place. Written as algebra, the six are: the carry a bijection  $s \mapsto (e, s - ed)$ ; the linear block's closed form  $[s_0 + rt]$ , the flight the translation group's shift along the digital line; the split's table with  $A = \sum a_i^2$  inverted by its conjugate transpose (Theorem 2); the rotation an integer matrix with  $U_s^\top U_s = n_s I$ ; the merge the addition in  $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ , its cancel the quotient by  $x^{N_\phi/2} + 1$ , which for  $N_\phi$  a power of two is the ring of cyclotomic integers  $\mathbb{Z}[\zeta_{N_\phi}]$ , a free  $\mathbb{Z}$ -module of rank  $N_\phi/2$ ; and the click one bilinear form  $\mathbf{f}^\top \mathbf{G} \mathbf{f}$  with  $\mathbf{G}$  the tables' Gram matrix, then one comparison. Modern algebra is not assumed here; it is what the cell rules are when written down, and every theorem below is a property of these six maps.

**The state.** The GameBoard is a box of Nodes in  $\mathbb{Z}^3$ , each joined to six neighbours by Links through six Ports; an axis may be periodic. Time is the interval  $t \in \mathbb{Z}$ . The state at an interval is a vector  $\mathbf{s}$  of bounded integers on a product of circles and lines, and it is the multiset of the messages in transit (the rows) and the bodies (the measured events), and nothing else: a Node holds no register, remainder or draw. A body carries its Node, its content  $M$ , its momentum vector  $\mathbf{p}$  and its table of counts, one accumulator per count (the fields per kind on the tree [10]).

**The components of Eq. (1).** The ladder's rungs  $b_k$  of Section 6 are walls of this kind. The components and their  $(r, d)$  [19]: a row's Manhattan count (the rate  $2S_1N_l$ , the wall  $2T_D$ , started at  $T_D$ ;  $D$  the direction vector,  $S_1$  its Manhattan length,  $T_D = \lfloor \sqrt{3|D|^2N_l^2} \rfloor$  its resolution,  $N_l$  the flight's grain), with three axis deficits choosing the axis of each Link; a row's phase (an integer per Link, or a declared rational  $\nu$  of steps per interval, written  $n/d$  in Section 6, its whole part read modulo  $N_\phi$ ); a row's age (the rate 1, no wall; bounded by the family's lifetime and the run); a body's drive on each axis (the rate  $|p_a|$ , the wall  $N_lN_wM + |p_a|$ ,  $M$  the content and  $N_w$  the world's width), whose speed  $p/(N_lN_wM + p)$  saturates at one Link per interval, above the rows' pace; a body's counts (the turn, the owed count, the release, the lamp, the push per column, each a rate against a wall); a record's birth phase (the rate 1 per birth, the wall  $N_\phi$ ; the birth wheel, a lamp's declared rate  $r_w$  per birth against the wall  $N_u$ , [2531, 4096] on the registered two-slit world, the rate the nearest odd integer to  $0.618 \times 4096$  [2]); the exact phase at the click[2]; and the click's ladder, one comparison read once. Beside the six operations stand the roundings declared at load, each a constant of the law, among them the tables of the circle at  $1/256$ , the unit vector of a direction to the nearest integer vector,  $T_D$  by the integer square root, the click's rung to the nearest integer and the fan's weights at the grain  $N_\theta$  [16].

**The two blocks.**  $\Phi$  is linear where its rates are constants of the world and feeds back where a rate is a function of what arrives. The linear block is the rows in flight: the flight, the phase and the age at constant rates, the merge, the integer matrices of a split, a rotation or a gate, and the evaluation at the click. Every component of it has the closed form of its unwrapped count,  $\lfloor s_0 + rt \rfloor$  (the accumulator itself is that count's remainder by the wall), so a row's state at any interval is a formula of its birth and its age, and the limits of Sections 4 and 5 are limits of that formula. The feedback block is the bodies: the push, whose rate is the reader's content times the flow of the arriving rows (bilinear in the state); the owed count, whose rate is what the body reads (the presence, or the age moment); the turn and the release, whose rate is the content; the drive, whose rate is the momentum the push changes. These have no closed form; they are iterated, and their continuum limit is the differential equation  $ds/dt = \mathbf{r}(\mathbf{s})$ . The exact word for the whole map is therefore piecewise-linear in the rows with bilinear rates in the bodies [19].

**The read-out.** The click reads one comparison and nothing else leaves the rows: for a record whose rows have all ended, the apparatus sums the rows' integer phasors per cell through the tables, squares the sums (the one quadratic step, forced in form by Theorem 4), lays the cells on a ladder and compares the record's coordinate  $u$  with the rungs; the cell whose rung  $u$  falls under is the click. Between a birth and a click every step of the linear block is injective on the rows' weights, multiplicities and phases for the event history the record keeps (Theorem 3); the click is the one deletion.

**A hand-worked update.** A row on the heading  $+x$  has the flight accumulator's rate  $2S_1N_l = 128$  ( $S_1 = 1$ ,  $N_l = 64$ ) against the wall  $2T_D$  with  $T_D = \lfloor \sqrt{3 \cdot 64^2} \rfloor = 110$ , so  $2T_D = 220$ , started at

$T_D = 110$ . Interval by interval the accumulator reads 238 (one wall counted: a Link, the remainder 18), 146 (none), 274 (a Link, 54), 182 (none), 310 (a Link, 90), 218 (none), 346 (a Link, 126), 254 (a Link, 34): Links at the intervals 1, 3, 5, 7, 8,  $\dots$ , five Links in eight intervals, and in the long run  $128/220 = 0.5818$  Links per interval, the heading's pace of Section 3. The same three lines of Eq. (1) run the phase (the rate the family's turn, the wall  $N_\phi$ ), a body's owed count and every other component; only the rate and the wall differ.

**In the code.** The engine realises Eq. (1) as one accumulator per component with its rate and wall read from the world file, the carry as the event, in six steps per interval: departures become arrivals (the walk), the readings at each Node, the collision, the measured events' tables and the detectors, the self-creations, and the border's lifetime with the merge of identical rows [2, 17]. The same engine runs every world of this paper from its world file.

**When the transition was made.** The road from ordinary physics to this algebra was walked in twenty-four dated steps, each a thing of physics becoming an object of the law, recorded with the reading that showed it [22] (Appendix C lists them): on 2026-09-17 the shared quantum resource was deleted and locality held without exception, the amplitude became a phase with a conserved content and Born's rule a declared table; on 09-18 the Born table was computed from  $N_\phi$  and a Node became its six Ports; on 09-19 a quantum became an integer row on a record, the click the one one-way border, the push one bilinear form and charge a rational per unit of content, and  $E = hf$  the cost of a release; on 09-20 every force became a column with a sign, the readings became two kinds, the masses the initialisation, every count an accumulator on its reader's own record, the Doppler the count of rows a mover crosses; on 09-21 the whole law became one vector operation of six verbs under three tests, the record an element of  $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$  with the weight one bilinear form, the lattice the translation group and  $c$  the norm of the flight operator, and the Lorentz factor, a root, was refused as a seventh verb; on 09-22 a detector's clock became a member of the age wall, the click theorem placed the Lorentz group Outside, and the paper's framing, Inside and Outside, was set.

**The ledger.** Table 1 is the paper's spine: for every result, the rules of Eq. (1) it starts from, the assumptions added beside them with their kind, the freedom left, and the ground of the derivation (the owner's decision of 2026-09-22 [18]): exact on the GameBoard (no limit, no average), a limit of the grain taken on paper with a rate, or a limit under a spatial condition, an average that covers the shell (over the shell's Nodes, over the directions of a fan declared in a ball, or over the sweep of an orbit), the condition then part of the claim. A result whose derivation is not closed, or whose only reading is of the GameBoard's state, is not in the table; the hypotheses beside the law are on the tree (Section 9).

Table 1: The forcing ledger: what the algebra shows, its ground word the rung; what a run measures is Table 2's. Ground: exact, on the GameBoard, no limit and no average; limit, of the grain, taken on paper with its rate; fan, a limit under an average that covers the shell, the condition part of the claim. A detector reading never lifts a claim from fan to exact: what a run establishes is a lattice-exact claim about the world at hand, whose agreement with the continuum's form within the named ripple is evidence for the fan claim, not its proof. Every row is derived from the update rules implemented in the simulator; the third column names what is added beside them with its kind; a result resting on a hypothesis beside the law has no row. The section is this paper's; the derivation's section is [19]'s.

| Result                                                                                                   | Rules it starts from                                     | Assumptions added (kind); freedom left                                                                                                                                                                                                         | Ground                                                      | Section     |
|----------------------------------------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-------------|
| The books' conservation; the continuity equation                                                         | the carry (a bijection), the split and rotation matrices | none                                                                                                                                                                                                                                           | exact (the lattice identity); limit (the differential form) | 4; 25.4     |
| The group of the six Ports, 48 and 24                                                                    | the six Ports, the hand bit                              | none                                                                                                                                                                                                                                           | exact                                                       | 3           |
| The pace of a direction $N_l D /T_D$ , isotropic within $1/T_D$ ; the Manhattan bound $S_1 N_l \leq T_D$ | one Link per interval, the digital line                  | none; the wall $T_D$ a rule chosen among few (P9); $N_l$ free                                                                                                                                                                                  | exact                                                       | 3; 13.2 (a) |
| The value $c = 1/\sqrt{3}$ attained                                                                      | the same                                                 | none                                                                                                                                                                                                                                           | limit (the grain $N_l$ )                                    | 3; 1.4      |
| No dispersion, the constancy of $c$                                                                      | the flight table indexed by direction and age            | the rate in transit constant, the flight blind to the phase (P9, a rule chosen among few), not forced by the six operations: a dispersive flight passes every requirement (the derivation's 27); (A1) of the frame asks the same of the clicks | exact                                                       | 3           |
| Gauss's law of a free family's flux                                                                      | the walk, the books                                      | a closed surface no periodic axis crosses, no absorber inside (a declared world condition)                                                                                                                                                     | exact                                                       | 4; 3.1      |
| The equivalence principle; the third law at rest                                                         | the step rule (divides by the content); the books        | none                                                                                                                                                                                                                                           | exact                                                       | 4; 3.3      |
| Newton's inverse square; $G = K\eta/(4\pi N_w)$                                                          | the bilinear push on the arriving rows' first moment     | two additivities (24.2); an average that covers the shell; $N_w$ an input                                                                                                                                                                      | fan                                                         | 4; 3.2, 3.3 |
| Coulomb's law with $k_C = G$                                                                             | the charge column of the push                            | $\rho$ an input; the same average                                                                                                                                                                                                              | fan                                                         | 4; 3.4      |
| The retarded potential and the retarded wave equation; Poisson's equation                                | the release, the flight, the age moment                  | the same average, the dwell direction-blind, the open box                                                                                                                                                                                      | fan                                                         | 5; 5.1, 5.5 |
| The Liénard-Wiechert potential of a moving source, first order                                           | the same                                                 | the same, the fan's grain                                                                                                                                                                                                                      | fan                                                         | 12.1        |
| The clock's rate $1/(1 + a_\tau)$ , never 0 at a finite count (no horizon)                               | the owed count                                           | the age moment as the clock's word (P9)                                                                                                                                                                                                        | exact                                                       | 5; 5.2      |
| The clock's $1/r$ form; its calibration                                                                  | the same                                                 | the $1/r$ form the same average; the calibration $a_\tau = GM/rc^2$ an input of the dictionary                                                                                                                                                 | fan (the form); input (the calibration)                     | 5; 5.2      |
| The click's weight a positive quadratic form of power 2                                                  | the evaluation at the roots of unity, the phase rotation | the balanced splitter's conservation (an axiom of the apparatus); the harmonic constants free; the least-rank click (P10)                                                                                                                      | exact                                                       | 6; 6.5, 6.6 |

| Result                                                                                                                                      | Rules                                  | Assumptions; freedom                                            | Ground                                                     | Section                               |
|---------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|-----------------------------------------------------------------|------------------------------------------------------------|---------------------------------------|
| Born's bound: a cell within $1/N_\phi$ , the cumulative within $1/(2N_\phi)$ ; Born's rule as its limit                                     | the ladder's rungs                     | the uniform birth phase (a fact of the source)                  | exact (the bound); limit (the rule)                        | 6; 6.2                                |
| The entropy identity; the entropy produced only at the click, the cancel and the leaks, a count of torus points; Boltzmann's $S = k \log W$ | the click's one deletion               | the coarse graining (a definition); no second law claimed       | exact                                                      | 14, 16.1                              |
| The split an isometry; the interval injective on the amplitudes for a fixed event history                                                   | the six operations                     | none                                                            | exact                                                      | 6                                     |
| Planck's $E = hf$ ; de Broglie's $\lambda = h/p$                                                                                            | the release's cost rule, the turn rule | the dictionary $h = h_q N_\phi = h_A$ , an input                | exact (identities under the dictionary)                    | 6; 6.4; 21.2 rows 51, 52; 24.1 row 25 |
| Young's spacing                                                                                                                             | two paths' phase difference            | the fan's grain                                                 | limit                                                      | 7.1                                   |
| The uncertainty relation's bounds on $\mathbb{Z}_{N_\phi}$                                                                                  | the evaluation                         | the circle as position, its transform as momentum: a hypothesis | exact (the bounds)                                         | (the 6; 22                            |
| Exact marginals; no-signalling                                                                                                              | the one gather, the nearest rung       | equal weights; open beyond                                      | exact                                                      | 7                                     |
| $S(N_\phi)$ exact, 181/64 on the plateau; $2\sqrt{2}$ the limit                                                                             | the rungs, the tables                  | the tables at $N_t = 256$ (P7)                                  | exact (the rational); limit (the bound)                    | 7; 24.4                               |
| The Doppler on the axis by the crossing rule                                                                                                | the crossing count, its boundary row   | none                                                            | exact (the count per Link); limit (the ratio $1 \pm v/c$ ) | 2.2, 2.7                              |
| A body's dispersion, its momentum's fraction; the moving clock's rate 1                                                                     | the drive per axis, the owed count     | none: the law's own predictions                                 | exact                                                      | 4.3, 4.4                              |
| Newton's cooling within one birth's cost                                                                                                    | the release                            | none                                                            | exact (a bound)                                            | 25.9                                  |

### 3 Geometry and symmetries: the directions and $c = 1/\sqrt{3}$

**The six Ports, the group and the front.** Everything geometric in the law follows from one Node and its six Ports,  $\pm x$ ,  $\pm y$ ,  $\pm z$ , and from one Link per interval. The maps of the six Ports onto themselves that keep opposite Ports opposite are the signed permutations of the three axes:  $3!$  orderings of the axes times  $2^3$  choices of sign, 48 in all, the symmetry group of the cube and of the octahedron (Theorem 1). The determinant of a signed permutation is  $\pm 1$ , so it splits the 48 into two halves of 24: the rotations, of determinant  $+1$ , which are the permutations of the cube's four body diagonals; and the improper maps, of determinant  $-1$ , the reflections with the central inversion among them. The hand of a row is that determinant made a column of the state: one pseudoscalar bit that a map  $g$  sends to  $\det(g)$  times itself, so a world with a hand reads which half a map belongs to. The same six Ports bound the pace: the Nodes one interval away are the six neighbours at the ends of the Links, the vertices of the octahedron  $|x| + |y| + |z| \leq 1$ , the causal front of one interval; a pace that is the same in every direction and crosses at most one Link per interval is at most the radius of that octahedron's inscribed sphere,  $1/\sqrt{3}$  Links per interval, the sphere touching the eight faces on the cube's diagonals (Proposition 1, Figure 1); that the rows' pace  $c$  sits at that bound and not below is the flight's declared wall  $T_D$ , the design's third statement and a rule chosen among few (P9), and given the wall the value is the flight operator's norm, attained exactly on the body diagonals (the derivation's 24.1, row 9).

**Theorem 1** (the 24 of the name). *The maps of the six Ports onto themselves that send opposite Ports to opposite Ports form the group of the signed permutations of the three axes, of order  $2^3 \cdot 3! = 48$ . The determinant is a homomorphism of this group onto  $\{+1, -1\}$ ; its kernel, the rotations of the cube, has index 2 and order 24, and is the group of the permutations of the cube's four body diagonals; the other coset, the reflections, has 24 elements too. The hand, the one pseudoscalar column of a row, which a map  $g$  sends to  $\det(g)$  times itself, tells the 24 rotations from the 24 reflections, as does an axial vector.*

The proof is Appendix A.

**The symmetries.** The law is invariant under the translations of the box and of the interval, and under the 48 signed permutations of the axes, the group of the octahedron of the six Ports and of the cube ( $3! \times 2^3$ ; the 24 of determinant  $+1$  its rotations, the 24 of  $-1$  its reflections, told apart by the hand, a pseudoscalar of the group), up to the law's two declared ties, the digital line's axis order and the collision's Port order ([2], note 39 and section 4); under the circle  $\mathbb{Z}_{N_\phi}$  exactly on the turns between events and at the click only up to the tables' rounding; and under the permutations of the families, each with its columns of the family table carried along [10]. A boost, which mixes a step in space with a step in time, is not among the 48: Lorentz's symmetry is not a symmetry of the GameBoard but of its clicks Outside, to second order in the velocity, the lattice's corrections from the fourth (Section 9). The 48 are the GameBoard's stand-in for the rotation group; angular momentum is conserved only up to the GameBoard's grain, and no Noether theorem for the map is claimed [6]. The name of the program counts them.

**The pace of the rows, and the octahedron.** Two rules are the whole kinematics of a row: it crosses at most one Link per interval (the locality of the GameBoard), and it follows the digital line of its direction, so that after one period of  $S_1 = |D_x| + |D_y| + |D_z|$  Links its displacement is the vector  $D$  itself (straightness). The Nodes one interval away from a Node are the six neighbours at the ends of its Links, the vertices of the octahedron  $|x| + |y| + |z| \leq 1$  (Figure 1); the octahedron scaled by the age is the causal front, inside which every row stays. A row's mean Euclidean pace lies within  $1/T_D$  of the octahedron's inscribed sphere (Proposition 1), so the vertex's pace, one Link per interval along a heading, is reached by no row's mean pace: the front is a bound and not a signal.

**Proposition 1** (The pace). *For every primitive direction  $D$ ,  $S_1 N_l \leq T_D$ : the Manhattan pace  $S_1 N_l / T_D$  is at most one Link per interval on every digital line, and  $S_1 = \sqrt{3} |D|$  exactly when  $|D_x| = |D_y| = |D_z|$ . A pace  $v$  that is the same in every direction and at most one Link per interval on every line satisfies  $v \leq |D| / S_1$  for every  $D$ , hence  $v \leq 1 / \sqrt{3}$ , the radius of the octahedron's inscribed sphere; and the table's Euclidean pace obeys  $1 / \sqrt{3} \leq N_l |D| / T_D < (1 / \sqrt{3})(1 + 1 / T_D)$  for every  $D$ .*

*Proof.*  $S_1^2 = (|D_x| + |D_y| + |D_z|)^2 \leq 3(D_x^2 + D_y^2 + D_z^2)$  is the Cauchy–Schwarz inequality against  $(1, 1, 1)$ , an equality exactly when the three sizes are equal; so  $S_1 N_l \leq \sqrt{3|D|^2 N_l^2}$ , and since  $S_1 N_l$  is an integer,  $S_1 N_l \leq \lfloor \sqrt{3|D|^2 N_l^2} \rfloor = T_D$ . A row at the Euclidean pace  $v$  on the line of  $D$  makes  $S_1$  Links per length  $|D|$ , so  $v S_1 / |D|$  Links per interval on average, and at most one Link per interval bounds this by 1:  $v \leq |D| / S_1$ . The least of  $|D| / S_1$  over directions is  $1 / \sqrt{3}$ , on the cube diagonals ( $1 / \sqrt{n}$  in  $n$  dimensions).  $T_D \leq \sqrt{3} N_l |D| < T_D + 1$  gives the pace's interval.  $\square$

So  $c = 1/\sqrt{3}$  Links per interval is the largest isotropic pace the two rules allow, and the table sits at it; that it sits at the supremum and not below is the flight's declared wall  $T_D = \lfloor \sqrt{3|D|^2 N_l^2} \rfloor$ , the design's third statement, a rule chosen among few (P9; the derivation's 24.1, rows 7 and 9), not a consequence of the two; given the wall the value is the flight operator's norm, so the value rests on locality, straightness and the declared wall. Stated exactly: the two rules give  $c \leq 1/\sqrt{3}$  (one Link per interval allows the pace 1 on a heading and only  $1/\sqrt{3}$  on a cube diagonal, three Links for the displacement  $\sqrt{3}$ ), and what is derived is the largest uniform pace the GameBoard allows under them; why the rows realise it, and what a detector in motion reads of it, are not derived here (Section 9). The anisotropy is the table's rounding, below  $1/T_D$  in every direction and so of order  $1/(\sqrt{3}N_l)$  (Table 3, row 5a, the bound on  $N_l$ ); nothing measured enters the value. The number is the bound of the Courant condition for the wave equation,  $c \Delta t / \Delta x \leq 1/\sqrt{n}$  [32], by this same argument, and the lattice Boltzmann sound speed  $c_s^2 = 1/3$  [33]; the paper adds the exact integer table at the bound and its rounding's anisotropy as a bound on the grain, and claims no novelty for the number. Stated with its ground: the pace of a direction,  $N_l|D|/T_D$ , isotropic within  $1/T_D$ , and the Manhattan bound  $S_1 N_l \leq T_D$  are exact in integers; the isotropic bound  $1/\sqrt{3}$  is a theorem about a pace the same in every direction, attained on the eight diagonals, above on every other registered direction, and a limit of the grain  $N_l$ .

**No dispersion.** What the GameBoard does not do to the pace is as definite as what it does.

**Remark 1** (The pace is the direction's). *By Definition 2 the flight table is indexed by  $(D, \tau)$ , the phase is a separate accumulator, and no rule of the interval (the flight, the birth, the split, the rotation, the merge; the archived collision table reads no phase) reads a phase into a direction or a step. So a row's pace is  $N_l|D|/T_D$  at every phase rate up to the alias bound of  $N_\phi/2$  steps per interval (a turn every two intervals, 1.15 to 1.16 Links), and the anisotropy of Proposition 1 is the same at every rate. This is the pure-shift case of a lattice automaton [56]: a ballistic walker with a passenger phase, free of dispersion by construction, as a classical corpuscle is. By P9 the pace has no dispersion at any phase rate, so the bound on an energy-dependent speed of light from GRB 090510 [34] is met, a consequence of the postulate and not a run. That no local linear wave scheme on the GameBoard is both isotropic and free of dispersion is Meyer's theorem [56], not this paper's.*

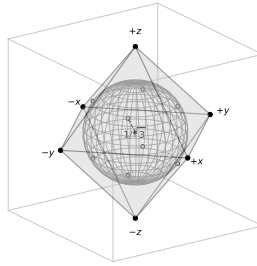


Figure 1: The Nodes one interval from a Node: the six neighbours are the vertices of the octahedron  $|x| + |y| + |z| \leq 1$ , the causal front of one interval; its inscribed sphere, of radius  $1/\sqrt{3}$ , touches the eight faces on the cube diagonals and is the rows' pace  $c$  (Proposition 1); the faint cube shares the 48 signed permutations of the axes. Drawn by `octahedron.py` from the definitions; no run.

**What a detector measures.** Two registered runs read the pace after a detector (Table 2): series Q, a lamp releasing one row on each of the 290 primitive directions with  $|a| + |b| + |c| \leq 6$  toward

detectors on the six faces, every click at the derived interval, Node and face, the pace 0.5718 to 0.5893 Links per interval at finite ages against the table's asymptotic 0.5774 to 0.5818; and series L7, the cone, a row of 17 Links on an axis and a row of 24 on the plane diagonal, the same Euclidean distance to one percent, at their counters at the same age.

## 4 Conservation, the flux and the forces

**The books.** Every message released is on exactly one line of the books at every interval, released equals in transit plus absorbed plus escaped plus cancelled, per family, in amount and in content, exact at every interval and at every operation (P3, P5): the carry is a bijection  $s \mapsto (e, s - ed)$  that loses nothing, and the split and the rotation are integer matrices whose conjugate transposes invert them (Theorem 2). From the books the continuity equation of the content follows exactly on the GameBoard (the derivation's 25.4 [19]); it is the first row of the ledger. A paid message carries its label out of its emitter at birth and into its reader at its end, so for paid messages momentum is conserved and the third law holds message by message; a free message, a field's row, takes no recoil at birth and pushes its reader by its label times the reader's content.

**The operations.** Six operations, of three of Section 2's kinds (the translation, the bilinear form, the Euclidean division), make the whole of gravitation and electricity in the law. The free release (a count against a wall): a body of content  $M_B$  holding a free family releases at each self-creation the carry of the count of rate  $M_B\eta$ ,  $\eta$  the release rate per unit of content per direction, against the wall 1 (the whole part of  $M_B\eta$  per self-creation,  $M_B\eta$  in the mean) on every declared direction, so a fan of  $K$  directions releases  $q = KM_B\eta$  per interval while the source's clock owes nothing. The walk (a translation): each unit walks its digital line at the pace of Proposition 1. The reading (a bilinear form): the label flow  $\mathbf{a} = \sum \text{amount} \times \mathbf{u}_D$  of the rows arriving at the reader's Node,  $\mathbf{u}_D$  the label per unit along the direction  $D$ , the integer unit vector at the scale  $N_l$ ,  $|\mathbf{u}_D| = N_l$  within 1.35 percent on every direction. The coupling (a bilinear form with a declared integer matrix): per axis push<sub>A</sub> =  $M_A(\rho_A\rho_B - 1)\mathbf{a}$ , the gravity column  $-M_A\mathbf{a}$  and the charge column  $+\rho_A\rho_B M_A\mathbf{a}$ ,  $\rho$  the declared charge per unit of content of each family. The momentum (a translation):  $\mathbf{p}_A \leftarrow \mathbf{p}_A + \text{push}_A$ . The step (the Euclidean division): one Link per  $d_p/|p_a|$  self-creations on the axis  $a$ , the wall  $d_p = N_l N_w M_A + |p_a|$ ,  $N_w$  the world's width; per interval, for a body whose clock owes nothing, the speed on the axis is  $|p_a|/(N_l N_w M_A + |p_a|)$ .

**Gauss's law, exact.** The walk is a translation: a free unit released inside a closed surface that no periodic axis crosses, with no absorbing reader inside it and within the unit's lifetime, crosses the surface once and never returns (no collision acts on a fan or on lone beams), so once the front has passed, the amount crossing the surface per interval equals the release inside it, exactly. Proved from the definitions under those conditions; no detector reading of it is registered.

**The inverse square as a shell mean.** On a beam the flow per Node is the same at every Node of the digital line: along a beam the field falls as  $1/r^0$ , and off every beam it is 0, by the walk. The inverse square is the density of beams over a shell. A shell at distance  $r$  holds  $\mathcal{N}(r)$  Nodes and the  $K$  beams cross it at  $K$  of them, so the mean over the shell of what a body reads is

$$\langle a \rangle(r) = \frac{q N_l}{\mathcal{N}(r)} \rightarrow \frac{q N_l}{2\pi r} \text{ on the plane, } \quad \frac{q N_l}{4\pi r^2} \text{ in space,} \quad (2)$$

as  $\mathcal{N}(r) \rightarrow 2\pi r$  and  $4\pi r^2$ ; the departure at finite  $r$  is the GameBoard's count of Nodes on a shell, Gauss's circle problem,  $\mathcal{N}(r) = 2\pi r + O(r^\theta)$  with the exponent  $\theta \leq 131/208$  [46], so the shell mean's relative ripple vanishes as  $r^{\theta-1}$ . No detector reading of the shell mean is registered;  $\mathcal{N}(r)$  is the GameBoard's count of Nodes on a ring, a number of the derivation and not a reading. Derived from the update rules implemented in the simulator, in the shell mean, with the ripple named; on one Node the inverse square exists only in the limit of every direction, and a fan declared as a cube of directions has in that limit a cubic anisotropy of the declaration,  $3^{3/2}$  between a corner and a face centre, absent for a fan declared within a ball.

**Newton's law and the place of  $G$ .** A body  $A$  of content  $M_A$  at the mean flow  $\langle a \rangle$  takes per interval the push  $-M_A \langle a \rangle$ , and its speed on an axis is  $|p_a|/(N_l N_w M_A + |p_a|)$ ; for  $|p_a| \ll N_l N_w M_A$  the acceleration is push/ $(N_l N_w M_A)$  Links per interval squared, so

$$a = -\frac{GM_B}{r^2}, \quad G = \frac{K\eta}{4\pi N_w} \quad \text{in space}; \quad a = -\frac{G'M_B}{r}, \quad G' = \frac{K\eta}{2\pi N_w} \quad \text{on the plane}, \quad (3)$$

in Links cubed (squared) per unit of content per interval squared:  $G$  is the source's release rate per unit of content per direction,  $\eta$ , times the number of directions  $K$ , over  $4\pi$  and the world's width  $N_w$ . No detector reading of either form is registered. Nothing of  $A$  enters: the equivalence principle is exact (the step rule divides by  $M_A$ ). The third law at rest is exact to the grain of the apportioning. Two departures from Newton's law as written, both stated: the field is retarded at  $c$  (the front at  $r$  arrives at the interval 1 plus the flight table's first age at  $r$  Links on the heading), so the limit is the retarded inverse square and not action at a distance; and the speed the push builds saturates at one Link per interval, above the rows' pace, so a body can outrun its own field's rows on the drive as built on main. Derived from the update rules in the shell mean,  $G$  named, the width  $N_w$  an input, a choice of units (Table 1).

**Coulomb's law and the one constant.** The charge column adds  $+\rho_A \rho_B M_A \mathbf{a}$  per axis. With the charges  $q_A = \rho_A M_A$  and  $q_B = \rho_B M_B$  and  $\langle a \rangle$  proportional to  $M_B$  as above, the electric push in the shell mean is

$$\text{push}_e = q_A q_B \frac{K\eta N_l}{4\pi r^2}, \quad a_e = \frac{q_A q_B}{M_A} \frac{G}{r^2}, \quad (4)$$

Coulomb's inverse square with the same constant as gravity,  $k_C = G$  in the law's units (charge in units of content), repulsive for like signs since  $\mathbf{a}$  points away from the source. The ratio of the two forces on one body is  $-\rho_A \rho_B = -q_A q_B / (M_A M_B)$  at every  $r$ . Derived under the same average as Newton's, the ratio  $-\rho_A \rho_B$  exact at every  $r$ ; no detector reading of it is registered. Nature's ratio for two protons,  $e^2 / (4\pi\epsilon_0 G m_p^2) = 1.24 \times 10^{36}$  ( $e$  the elementary charge,  $\epsilon_0$  the vacuum permittivity,  $m_p$  the proton's mass), is  $\rho_p^2$  here,  $\rho_p = 1.1 \times 10^{18}$  units of charge per unit of content: the hierarchy is the declared  $\rho$ , an input, not a formula of the law.

**The ground of these derivations.** Gauss's law of the flux, the equivalence principle and the third law at rest are exact on the GameBoard. The inverse square, Newton's and Coulomb's, is a limit under a spatial condition, an average that covers the shell, in one of three admissible forms, each with its own error: over the shell's Nodes (the shell mean, the ripple  $O(r^{\theta-1})$  with  $\theta \leq 131/208$ , at any declared fan); over the directions at one Node, the fan dense at fixed  $r$ , and then only for a fan declared in a ball (a cube's fan keeps a cubic anisotropy of  $3^{3/2}$  in the limit); or over the path of a body that sweeps the fan's lines, one closed turn of an orbit, whose mean push per turn is the ring mean exactly, the only one of the three readable after a detector. On one Node beside a source

the field is a comb of beams, and no single reading of the GameBoard's state is the inverse square; the condition is part of the claim (the ledger's caption). After a detector the equivalence principle is measured (series D3, Table 2) and the  $1/r$  force's scale symmetry reads consistent,  $1.997$  for  $2.00 \pm 0.18$ , on loops that are not similar figures: the inverse square is not closed after a detector and  $G$ 's value is not read; Coulomb's and Poisson's runs are not made (Section 9).

## 5 The delay field: the retarded potential, Poisson's equation and the clock

The rows of Table 1 on the delay field are written out here from the derivation's section 5 [19]; the register's series E reads probes on the GameBoard, diagnostics this paper does not carry, and the field's one measurement after a detector is the clock's form at two distances, series T (row 12 of Table 3).

**What is delayed.** Nothing in flight is delayed: a row moves by the flight table at one pace, blind to the crowd (series K, Table 2). What a crowd delays is a clock. A body reads at its Node the presence of every row of another number (the engine's default) or, on an entry that reads the age, their age moment (the clock's word, P9) and owes after each self-creation the whole part of  $a_r n_c/d_c$  intervals before the next,  $a_r$  what it read and  $[n_c, d_c]$  the clock's pair of Section 2 ( $[1, 1]$  in the scalar world,  $[1, 2]$  in the age world), so that its rate is  $1/(1 + a_r n_c/d_c)$ ; its push reads the flow of the same rows.

**The two fields of one stream.** Let a source release  $q(t)$  rows per interval on a fan of every direction (the limit of Section 4's shell mean), each row moving at  $c$  and carrying its age. The presence  $\Pi$  at the distance  $r = |\mathbf{x}|$  from the source at the time  $t$  is the rows dwelling there, and the age moment  $\mathcal{A}$  is the presence times the age they carry,  $r/c$ :

$$\Pi(\mathbf{x}, t) = \frac{q(t - r/c) \tau_L}{4\pi r^2}, \quad \mathcal{A}(\mathbf{x}, t) = \frac{q(t - r/c) \tau_L}{4\pi c r}, \quad (5)$$

$\tau_L$  the radial residence factor, the intervals a row spends per unit of Euclidean distance,  $T_D/(N_l|D|) = 1/v_D$  ( $1.72$  on a heading,  $\sqrt{3}$  in the limit of every direction); the intervals per lattice Link along a row's staircase are  $T_D/(S_1 N_l)$ , a different factor ( $1$  on a body diagonal), and a finite shell's coefficient depends on its digital geometry, so the coefficient of (5) is the heading's on a beam and the limit's on the sphere. The presence is the retarded flux of a conserved stream; the age moment is the retarded potential of the source, since  $f(t - r/c)/(4\pi r)$  is the retarded Green's function of the wave operator, so  $\mathcal{A}$  obeys the inhomogeneous scalar wave equation with the release as its source,

$$\frac{1}{c^2} \frac{\partial^2 \mathcal{A}}{\partial t^2} - \nabla^2 \mathcal{A} = \frac{\tau_L}{c} q(t) \delta^3(\mathbf{x}), \quad (6)$$

$\delta^3$  the three-dimensional Dirac delta, and in the static limit Poisson's equation  $\nabla^2 \mathcal{A} = -(\tau_L/c) q \delta^3(\mathbf{x})$ , solved by  $\mathcal{A} = (\tau_L q / 4\pi c) / r$ . The moments are sums over rows, so the field of many sources is the sum of their fields: the theory is linear. The flow the push reads is the stream's flux vector,  $\mathbf{a} = (q N_l / 4\pi r^2) \hat{\mathbf{r}}$ ,  $\hat{\mathbf{r}}$  the unit vector from the source and  $N_l$  the label each unit carries, which is  $-(N_l c / \tau_L) \nabla \mathcal{A}$  in the static case: the acceleration of a body is minus the gradient of the age moment's field. Derived from the update rules, in the limit of every direction: the age moment is the retarded potential of the release and obeys the wave equation there. No detector reading of the two fields is registered.

**The clock's redshift.** A clock at the age moment  $a_\tau = C_a M/r$ ,  $C_a$  the constant of Eq. (5) times the clock's pair, runs at  $1/(1 + a_\tau)$ ; general relativity's clock at the potential runs at  $\sqrt{1 - 2GM/rc^2} = 1 - GM/rc^2 - \frac{1}{2}(GM/rc^2)^2 - \dots$ . Under the calibration  $a_\tau = GM/rc^2$ , an input of the dictionary (by the derivation's 5.1 and 3.3 the clock's constant and Newton's  $G$  have the exact ratio of the suspension pair times the width, so the calibration is a constraint on the inputs), the two agree at first order: the redshift between two clocks,  $1 - a_\tau(r_1) + a_\tau(r_2)$ , is the  $1/r$  law. Derived at first order, the calibration an input. At second order the GameBoard reads  $1 - a_\tau + a_\tau^2$  against  $1 - a_\tau - a_\tau^2/2$ , and its rate never reaches zero at a finite  $a_\tau$ : a different law in the strong field, no horizon (the crowd only slows; nothing stops a clock). No detector reading of the strong-field form is registered; the weak-field confrontation ( $a_\tau \ll 1$ ) is a run the register names as the follow-up, neither pinned nor made. Which word the clock counts, the presence or the age moment, was put to the GameBoard as series T, the physicist's pins written before the run [4, 13]: a lamp at rest, two sources at 3 and at 6 Links, the detector at  $x = 110$ ; under the presence word the detector reads  $1 + z = 1.3000$  at both distances; under the age word 2.6517 and 4.1500, the ratio of the two  $a_\tau$  being 1.907 for the lines' pin 1.909, nature's 2.00 being 4.6 percent away; every pin met, measured and replicated [9]. What the detector establishes is the form at two Nodes, a lattice-exact reading; the continuum's  $1/r$  is its limit under the average that covers the shell, and the reading is evidence for it, not its proof. The age moment is the clock's word, an assumption of the law (P9; record 394 [6]), not a derivation; the law as built counts the presence by default and the age moment on an entry that reads the age; the presence word is the alternative refuted on the form.

**Light: no optical metric.** A row reads nothing of the crowd: beside a mass it is neither bent nor delayed, exactly (series K, three worlds: the deflection 0.000 pixel and the delay 0.00 interval at a crowd where nature would capture the beam, against nature's 1.75 arcseconds at the Sun's limb; the derivation's 24.3, row 14, REFUTED on main; Table 3). The equivalence principle holds for bodies and not for rows, and the delay field is a metric for clocks and for the bodies' pushes, not for the flight.

## 6 Measurement and the quadratic form

The worlds of this section have no body: the law of Section 2 on rows alone, and the measurement is the one threshold that is read out. Every interval applies three of the six operations to the state, the translation (the flight, the phase, every count and the birth), the multiplication by a declared integer matrix (the split and the rotation) and the sum in the group ring (the merge, where opposite phases cancel); between clicks the rates are constant, so the map has a closed form and every count of Section 8 was computed from it before its run. The click is the one place with a fourth operation: the apparatus sums a record's rows through the tables and squares the sum, the norm, and only then applies its threshold to the birth phase  $u$ .

**Definition 1** (Row and record). A *row* is  $(x, D, \tau, p, e, h, r, \ell, w, \mathfrak{m})$ : Node, direction, age, phase  $p \in \mathbb{Z}_{N_\phi}$ , emitter, content per unit, record  $r$ , branch label  $\ell$ , amount  $w \geq 1$ , multiplicity  $\mathfrak{m} \geq 1$ . Its amplitude is

$$z = \frac{w}{\sqrt{\mathfrak{m}}} \frac{C_{N_\phi}[p] + iS_{N_\phi}[p]}{256}, \quad |z|^2 = \frac{w^2}{m} \cdot \frac{C_{N_\phi}[p]^2 + S_{N_\phi}[p]^2}{65536}. \quad (7)$$

Two rows equal in every field but  $w$  are one row with the amounts added; two rows of one record and label equal in every field but a phase difference of  $N_\phi/2$  are one row with the amounts subtracted

(an equal pair is no row). The state of record  $r$  is an element  $\psi_r$  of the free  $\mathbb{Z}$ -module  $M_r$  on the basis  $(x, D, \tau, p \bmod N_\phi/2, e, h, \ell, \mathbf{m})$  with  $|p + N_\phi/2\rangle = -|p\rangle$ , taken modulo the scaling  $(w, \mathbf{m}) \sim (kw, k^2\mathbf{m})$ ; the GameBoard's state is one normal form per element of  $\bigoplus_r M_r$ . The norm is  $\langle\psi, \psi\rangle = \sum_{\text{rows}} w^2/\mathbf{m}$ . The archived engine narrows the quotient at a detector: rows of one record with different  $\mathbf{m}$  at one set are added when the ratio of the multiplicities is the square of a rational (the pointers rescaled by the integer roots) and refused otherwise; at the tree of the runs two multiplicities at one set were refused, and no reported world has them, so every set below reads one  $\mathbf{m}$ .

The host, not any Node, keeps per live record the birth phase  $u \in \mathbb{Z}_{N_\phi}$ , the count of live units (amounts), an accumulated pointer  $(X, Y)$  per (Node, label) and a gathered flag. This is the apparatus's layer; the GameBoard never reads it. It is a memory no Node could own: on the two-slit world the two paths of one record reach a pixel up to 18 intervals apart, so no Node ever holds both rows [6].

**Definition 2** (The update rules of one interval). Applied in this order to every row.

- F Flight:  $(x, D, \tau, p) \mapsto (x + \text{pos}_d(\tau + 1) - \text{pos}_D(\tau), d, \tau + 1, p + \varphi(\tau))$ , where the family's phase turn is either an integer per Link, added only when the step is a Link, or a pair  $[n, d]$  per interval of age,  $\varphi(\tau) = \lfloor(\tau + 1)n/d\rfloor - \lfloor\tau n/d\rfloor$ , counted from the row's birth or last re-emission.
- C Collision: at a Node holding several rows of amount 1 the table permutes their directions as a function of the whole set present;  $r, \ell, \mathbf{m}$  unchanged; a bijection of states, not a linear map. No collision acts in any world of this paper.
- S Split, at a measured event with outputs  $\{(d_i, a_i, t_i)\}$ ,  $a_i \in \mathbb{Z}$ : the row  $(w, \mathbf{m}, p)$  is absorbed and re-emitted as the rows  $(wa_i, \mathbf{m}A, p + t_i)$  on  $d_i$  at age 0,  $A = \sum_i a_i^2$ . The balanced splitter is  $(1, 1)$ ,  $A = 2$ ; the reflection's turn is  $N_\phi/4$ .
- R Rotation, at a set with setting  $s$  and turn  $t$ , on a label pair  $(0, 1)$  read into the channels  $(+, -)$ :  $U_{s,t} = \begin{pmatrix} C'[s] & S'[s]v(t) \\ -S'[s] & C'[s]v(t) \end{pmatrix}$  with  $v(t)$  the turn by  $t$  steps on the label-1 column; a row of label 0 becomes  $(wC'[s], 65536\mathbf{m}, p)$  on  $+$  and  $(wS'[s], 65536\mathbf{m}, p + N_\phi/2)$  on  $-$ , a row of label 1 becomes  $(wS'[s], 65536\mathbf{m}, p + t)$  on  $+$  and  $(wC'[s], 65536\mathbf{m}, p + t)$  on  $-$  (the row form exact for  $t$  a multiple of  $N_\phi/4$ , the only turns used here; the archived code multiplies the column by the tables' phasor at  $t$ ). With  $t = 0$  this is  $U_s$  of Theorem 2; the GHZ bases below are  $X = (s, 0)$  and  $Y = (s, N_\phi/4)$ .
- M Normal form: merge and cancel as in Definition 1.
- B Birth: a lamp's self-creation at interval  $t$  releases record  $r$  with  $u$  the lamp's phase at the birth,  $(b-1) \bmod N_\phi$  for a lamp started at 0 that turns one step per birth,  $b$  its count of births (every world here;  $b-1 = t-1$  only while the lamp births at every interval; the archived code writes  $(b-1) \bmod N_\phi$  directly), rows of amount 1 on its  $k$  directions with  $\mathbf{m} = k$ , each row's phase starting at  $u$  (plus the lamp's own turn), and, for a pair, the declared labels with integer weights on separate arms.

**Definition 3** (The click). At a set that reads the sum, the rows of record  $r$  and label  $\ell$  that arrive are absorbed and the layer accumulates

$$X += \sum 32 w C_{N_\phi}[p], \quad Y += \sum 32 w S_{N_\phi}[p], \quad (8)$$

the phase  $p$  as the row carries it on arrival (the turns of rule F already in it). The pointer is kept per Node: interference is coherent within one Node only, and the offer of a set of Nodes (a detector, a face) and label  $\ell$  is the sum over its Nodes of  $(X_x^2 + Y_x^2)/\mathfrak{m}$ , one cell of the ladder, the click landing at the Node within the set that  $u$  selects by the same rungs; all rows at a set share  $\mathfrak{m}$ . When the record's live count reaches 0, with  $C_K = \sum R$  over its offers and the cumulative sums  $C_1 \leq \dots \leq C_K$  (the total) in the declared order,

$$b_k = \left\lfloor \frac{2N_\phi C_k + C_K}{2C_K} \right\rfloor \quad (b_0 = 0, \ b_K = N_\phi), \quad \text{the click is the } k \text{ with } b_{k-1} \leq u < b_k, \quad (9)$$

and the world's list gains one row: the record,  $u$ , the chosen set and label,  $R$ ,  $C_K$ , the count of combinations before and after. For a record with arms at two rotated sets  $A$  and  $B$  (settings  $a, b$ ) the cells are the joint outcomes  $(o_A, o_B)$  in the arms' declared order, with the weights

$$R(o_A, o_B) = J(o_A, o_B)^2, \quad J(o_A, o_B) = \sum_{\ell} c_{\ell} U_a[o_A][\ell] U_b[o_B][\ell], \quad (10)$$

$c_{\ell}$  the labels' declared weights,  $(1, 1)$  in every world here; and the record is gathered once, when the rows at both arms have ended: the world's row carries both outcomes at the completion interval, so the near party's outcome is written at the far party's tick. Reading the ladder in that order is the same as a coarse rung over  $o_A$  followed by a fine rung over  $o_B$  within it with the same  $u$ , which is how Theorem 5 states it. A set that reads without absorbing (the which-path read of the polariser) stores the record's residual selector keyed by its labels and lets the rows go on; it is a deferred offer, gathered at the record's one click on the labels present then, not an outcome at the interval of the read. A read, then a rotation of the labels, then a second read, taken as two outcomes in time, is not modelled: the first read's selector is applied once, at the gather, to the labels as the rotation left them [17].

**Lemma 1** (One click per record). *A record every row of which ends (at a set, a face or the border; not a row parked at rest or on a periodic axis without a lifetime) completes and is gathered exactly once: the live count reaches 0;  $b_K = N_\phi$  (with  $C_K > 0$ , the born row's norm up to the tables' rounding, Theorem 2; at  $C_K = 0$  the engine gathers nowhere, record 363 [6]) and  $b_0 = 0$ , so every  $u \in \mathbb{Z}_{N_\phi}$  lies in exactly one cell. Every record of the worlds below is of this kind; the delay between a record's last arrival and its completion is the record's, not the GameBoard's.*

Three things are put in here and not derived, and the paper claims none of them. Definition 3 recovers the Born rule's form: the weight is the squared length of the summed pointer, and the ruler over  $u$  is inverse-transform sampling of a uniform variable, so over births with  $u$  uniform on  $\mathbb{Z}_{N_\phi}$  the count in channel  $k$  is  $b_k - b_{k-1}$ , the difference of two cumulative rungs, Born to the resolution  $1/N_\phi$  per cell. No principle below the square is offered. The uniformity of  $u$  is an assumption about the source:  $u = (b - 1) \bmod N_\phi$  is uniform over  $N_\phi$  consecutive births, (one birth per  $u$  in every run below), and its independence of the settings is a fact of a world, not of the model:  $u$  is a clock and a chooser is a clock, so they are independent when the chooser's period is coprime to  $N_\phi$  and the run covers their common period, as in this paper's choosers' world (periods 3 and 5 against  $N_\phi = 64$ ). The tables at  $1/256$  are the third of these three, and a declared input of the law (Section 2, P7). What the paper derives is what follows from these with the GameBoard's rules: the interference of one record, the exact marginals, the  $\cos^2$  correlation of the pair, GHZ's zeros, and the finite- $N_\phi$  value of  $S$ . The click is the only step that removes combinations and adds a row to the world's list; the birth is the only step that adds combinations.

**The two theorems of the interval.** Two theorems of the linear block are stated here and proved in Appendix A.

**Theorem 2** (The split is an isometry whose conjugate transpose inverts it). *The split  $(w, \mathbf{m}, p) \mapsto ((wa_i, \mathbf{m}A, p + t_i))_i$  preserves  $\sum w^2/\mathbf{m}$  for every nonzero integer vector  $(a_i)$ , and the conjugate transpose of the table (the same weights with the turns reversed) applied to its outputs, followed by the normal form, returns the input up to the scaling  $(kw, k^2\mathbf{m})$  with  $k = A$ . The rotation satisfies  $U_s^\top U_s = n_s I$  exactly with  $n_s = C'[s]^2 + S'[s]^2$ , and multiplies a record's norm by  $n_s/65536$ : an isometry only up to the tables' rounding ( $65705/65536$  at  $N_\phi = 64$ ,  $s = 1$ ).*

**Theorem 3** (The interval is injective between clicks). *Let  $Q$  be the quotient of  $\bigoplus_r M_r$  that forgets the age (rows differing only in their age identified), and fix the event history of the record, the intervals and Nodes of its births, splits, rotations and re-emissions, which the record keeps. For an interval without a click, an end at a face or the border, or a collision,  $M \circ R \circ S \circ F$  induces an injective  $\mathbb{Z}$ -linear map of  $Q$  into itself: no rule of the interval deletes a weight, a multiplicity or a phase, and the state at any interval between two clicks, with the history, determines them at the birth. The age is not a coordinate of  $Q$  but a function of the history, the current interval less the interval of the row's last event; a split restarts it at 0 by the definition of age (Definition 2) and forgets nothing, since the interval of the split is in the history; the click is the one deletion. A collision, where it acts, is a bijection of the GameBoard's states that is not linear; none acts in the worlds of this paper.*

So a record's norm is conserved exactly by  $S$  and by the flight, and by  $R$  and by every read through the tables only within the tables' rounding (Appendix B). The normalisation by  $C_K$  in (9) makes the click's probabilities sum to one at completion; it does not restore the norm along the way, and the paper claims exact unitarity nowhere.

**The click's square as a positive quadratic form: a lattice Gleason.** The row of Table 1 on the click's weight is written out here from the derivation's section 6.5 [19]; the register's checks are series L's clicks (Section 8). The click's square, the norm of a record's summed pointer, is the one operation of the law beyond the translation, the wall, the declared matrix and the addition, and the place where the Born rule enters. Asked: whether the operations of Section 2, with one hypothesis on the reading, force the reading at a click to be that square, or what weaker statement holds and what the register pins.

**The objects.** A record's rows at one Node and label are one element  $f = \sum_p f_p x^p$  of the group ring  $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ ,  $x$  the generator of one phase step and  $f_p$  the amount at the phase  $p$ ; the pointer is  $\text{ev}(f) = \sum_p f_p \zeta^p$ ,  $\zeta$  the primitive  $N_\phi$ -th root of unity, at the tables' scale. The merge's cancel identifies  $x^{N_\phi/2}$  with  $-1$ , so the rows after the cancel are an element of  $\mathbb{Z}[x]/(x^{N_\phi/2} + 1)$ ; for  $N_\phi$  a power of two (every registered  $N_\phi$ ) that quotient is the ring of cyclotomic integers  $\mathbb{Z}[\zeta_{N_\phi}]$  itself, a free  $\mathbb{Z}$ -module of rank  $N_\phi/2$ , and  $\text{ev}$  is the identity on it: the GameBoard's cancel and the click's zero are one relation, exactly (for an  $N_\phi$  with an odd factor they are not; the register has no such  $N_\phi$ ). The reading is a function  $R : \mathbb{Z}[\zeta_{N_\phi}] \rightarrow \mathbb{R}$  of a cell's element, up to one factor per record, which the multiplicity  $\mathbf{m}$  books.

**The hypotheses.** (a) The phase rotation,  $R(xf) = R(f)$ : a rotation of the record's phase by one step changes no weight (the rungs take no  $u$ ;  $u$  only selects among them), an operation of Section 2. (b) The balanced splitter conserves the reading for every two inputs,  $R(x^{N_\phi/4}f + g) + R(f + x^{N_\phi/4}g) =$

$A_2(R(f) + R(g))$  for all  $f, g$  (the (1, 1) split with the quarter turn on the reflected row,  $A_2$  its multiplicity factor): a hypothesis on the reading, stronger than Section 2's split, which is an integer matrix on rows and says nothing of what a detector reads under it; the register checks it only on the registered pairs (the totals 64/0 and 32/32 of Section 8). (c) Counts,  $R(f) \geq 0$ . (d) The empty reads zero:  $R(0) = 0$  on the zero count vector. (e) Some input reads:  $R$  is not identically zero. Without (d) the hypotheses admit a detector that always reads 1, which satisfies (a), (b) with  $A_2 = 1$  and (c) and is no quadratic form; without (e) they admit the zero detector, under which (b) holds for every  $A_2$ ; so (a) to (c) alone do not force the square. (d) and (e) are what the engine does (the Gram form on the empty record is 0; a click needs a live unit) and what the proof's first step uses (the model owner's finding, record 340 [6]; the derivation's 6.5 restated with them). Not assumed: continuity, a dimension, the form of  $R$  on one row, or Theorem 2's  $A = \sum a_i^2$ .

**Theorem 4** (The lattice Gleason). *Under (a) to (e), with  $N_\phi$  a power of two,  $A_2 = 2$  and*

$$R(f) = \sum_{j \text{ odd}, 1 \leq j < N_\phi/2} c_j |\sigma_j(f)|^2, \quad c_j \geq 0, \quad \sigma_j(f) = \sum_p f_p e^{2\pi i j p / N_\phi}, \quad (11)$$

*the  $\sigma_j$  the Galois conjugates of  $\text{ev}$ ,  $\sigma_1 = \text{ev}$ . Every such  $R$  is a positive quadratic form on the lattice, homogeneous,  $R(af) = a^2 R(f)$ ; it conserves the total at every declared split of one row or of two recombining inputs exactly if and only if the multiplicity factor is  $A = \sum a_i^2$ , and at every label rotation with the factor  $C'^2 + S'^2$ ,  $C'$  and  $S'$  the half-angle tables of Definition 2. Two rows  $ax^p$  and  $bx^{p+\Delta}$  at one Node,  $\Delta$  their phase difference in steps, read  $R = C_\Sigma(a^2 + b^2) + 2ab\mathcal{K}(\Delta)$  with  $C_\Sigma = \sum c_j$  and  $\mathcal{K}(\Delta) = \sum c_j \cos(2\pi j \Delta / N_\phi)$ ,  $\mathcal{K}(0) = C_\Sigma$ ,  $\mathcal{K}(N_\phi/4) = 0$ ,  $\mathcal{K}(N_\phi/2) = -C_\Sigma$  for every admissible  $R$ ; the Born form  $|\text{ev}(f)|^2$  is  $c_1 = 1$  and the other  $c_j = 0$ .*

*Proof.* Sketch; the derivation's 6.5 [19] gives the steps in full under (a) to (e). (i)  $g = 0$  in (b) with (a) gives  $2R(f) = A_2(R(f) + R(0)) = A_2 R(f)$  by (d), so  $A_2 = 2$ , one  $f$  with  $R(f) > 0$  sufficing, and (e) supplies it. (ii) Replacing  $g$  by  $x^{-N_\phi/4}g$  in (b) and using (a) and  $x^{-N_\phi/2} = -1$  gives  $R(f - g) + R(f + g) = 2R(f) + 2R(g)$ , the parallelogram law on the abelian group  $\mathbb{Z}[\zeta_{N_\phi}]$ . (iii) The parallelogram law alone makes  $R$  a quadratic form: on a lattice the argument of Jordan and von Neumann [54] needs no continuity, and  $R(f) = B(f, f)/2$  with  $B$  a  $\mathbb{Z}$ -bilinear symmetric form, the restriction of the real form of a symmetric matrix  $\mathbf{G}$ , the click's Gram matrix (not the fan's grain  $N_\theta$  of Section 2). (iv) By (a),  $\mathbf{G}$  commutes with the signed permutation of the basis that multiplication by  $x$  is, whose eigen-functionals are the  $\sigma_j$ ,  $j$  odd; so  $\mathbf{G}$  is diagonal in that basis with real entries paired under  $j \leftrightarrow -j$ . (v) A negative  $c_j$  would make the form negative on an open cone containing lattice points, against (c). (vi) The conservation at a general split and at a label rotation follows from homogeneity, (a) and the cancellation of the cross terms between the two outputs. (vii) Two rows:  $|\sigma_j(a + bx^\Delta)|^2 = a^2 + b^2 + 2ab \cos(2\pi j \Delta / N_\phi)$ .  $\square$

**What is derived and what is not.** Derived from the update rules and the axiom (b): the reading is a positive quadratic form of the record's element, forced by the rotation, the balanced splitter's conservation and the counts alone, with no continuity and no dimension (Gleason's theorem [47] needs dimension three; the lattice version does not, because (b) holds for every two inputs, the parallelogram law, a stronger hypothesis the balanced splitter grants). The multiplicity rule  $A = \sum a_i^2$  is derived here, not assumed; the square is the only power ( $R(af) = a^2 R(f)$ : no  $|w|^k$  with  $k \neq 2$  conserves); and the cross term  $2ab\mathcal{K}(\Delta)$ , a product between two rows the GameBoard never forms, is forced to exist at the click, because the two outputs of a balanced splitter conserve the total only through cross terms that cancel between them. Not reached by the algebra: the constants  $c_j$ , the detector's response to the harmonics of the phase, the exact analogue of Gleason's

free state: a free positive operator commuting with the rotation, diagonal on the Galois planes. A single harmonic  $j$  alone is the Born form under the relabelling  $p \rightarrow jp$  of the phase steps, a change of the declaration and not of the law; a mixture is a different law, the flat mixture being the phase-blind detector, which satisfies (a), (b), (c) exactly; what excludes it is interference, nature's (Table 3, row 2a) and, for the implementation, the built click's.

**Against the registered integers.** The built click carries the square: the  $(3, 4)$  split's  $63/1$ ,  $31/1$  and  $125/3$  at  $N_\phi = 64$ ,  $32$  and  $128$  (record 363 [6]) hold if and only if the power  $k$  lies in  $[1.917, 2.012)$ , containing 2 and excluding 1 and 3, a check that the implementation is the form of Theorem 4 and no evidence about nature's power (Table 3, row 2c); the norm range of Appendix B is the error bound between the ideal reading and the tables. The two-slit period 23.5 pixels and Malus at 22.5 degrees (219/256 exact) admit the harmonic  $j = \pm 1 \bmod 8$  and exclude every mixture: the register pins the fundamental, and the least-rank click (P10) stores it.

**Interference, by the formulas alone.** Every phenomenon of this paper is read the same way: Outside, what a detector counts; down to Inside, the rows, their phases, the merge and the evaluation; back Outside, the click's counts over the births against the reading. Two slits: one record is born with two rows of amount 1, one per opening,  $m = 2$ ; each row's phase advances one turn per Link; at pixel  $x$  the two arrive with phases  $p_1(x)$ ,  $p_2(x)$ , their difference  $\Delta(x)$  the path difference times the turn; the merge adds them and an equal pair with  $\Delta = N_\phi/2$  cancels to no row; the click evaluates the sum and squares it, the offer  $R(x) = 512 [(C[p_1] + C[p_2])^2 + (S[p_1] + S[p_2])^2] = 1024 (65536 + C[p_1]C[p_2] + S[p_1]S[p_2])$  up to the tables' rounding, proportional to  $1 + \cos \Delta(x)$ ; the rungs  $b_x$  of Eq. (9) lay the pixels on the ladder, the click is the pixel under which  $u$  falls, and over  $N_\phi$  births the count at  $x$  is  $b_x - b_{x-1}$ , Born's to  $1/N_\phi$ . What passes: the pixels with  $\Delta$  near 0, a wide rung. What does not: the pixels with  $\Delta = N_\phi/2$ , the rows cancelled, the offer 0, a rung of width 0 where no click can land. One slit: one row per pixel, no pair, nothing cancels, a flat count within the fan's grain, no dark pixel. Mach-Zehnder: the balanced splitter, two arms, the mirrors' turn  $N_\phi/4$ , the second splitter; at one port the two phasors add and at the other they are  $N_\phi/2$  apart and cancel; with the tables at  $N_\phi = 64$  the offers are 1681/1682 and 1/1682, the rounding and not 1 and 0, and the rungs give the bright port 64 of 64 clicks and the dark port 0 (row 2b); the half turn and the quarter turn, 0/64 and 32/32, follow by the same lines. The pair (Section 7), Malus's 219/256 at 22.5 degrees from the rotation table, and GHZ's zeros from the joint weights are the same three lines each; the register's runs confirm them (Section 8).

**The limit.** As  $N_\phi = 2^k$  grows the admissible readings are the same family with  $N_\phi/4$  constants: for two rows at the angle  $\theta = 2\pi\Delta/N_\phi$ ,  $\mathcal{K}(\theta) = \sum_{j \text{ odd}} c_j \cos(j\theta)$ , the closed cone of the even positive-definite functions on the circle with odd harmonics only, whose extreme rays are the single cosines; the Born kernel  $\cos \theta$  is the first of them. The limit adds nothing: no  $N_\phi$  and no continuity forces  $c_j = [j = 1]$ ; the fundamental is what the register reads and what the phase means.

**Planck and de Broglie, as identities.** Under the declared dictionary the release's cost rule reads as  $E = h_q s = (h_q N_\phi) f$  and the turn rule as  $\lambda = h_A/p$ : Planck's and de Broglie's relations are identities of the update rules with one constant,  $h = h_q N_\phi = h_A$ , an input (Table 1); no detector reading of them is registered.

**The uncertainty relation, from the same evaluation.** A record is the element  $f$  of  $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$  of the amounts that ended at each phase. The click evaluates  $f$  at the  $N_\phi$ th roots of unity, which is

the discrete Fourier transform on the circle. That the circle is read as position (the Node its rows ended at) and its transform as momentum (the label whose turn per Link is  $|p|N_\phi/h$ , de Broglie's  $\lambda = h/|p|$  Links as the declared turn) is a stated identification of the program, a hypothesis and not a derivation ([19], section 22); on it, three statements are the transform's mathematics on  $\mathbb{Z}_{N_\phi}$  and nothing of the law. The support bound: a record whose rows end on  $s$  phases has an evaluation supported on at least  $N_\phi/s$  roots,  $|\text{supp } f| |\text{supp } \hat{f}| \geq N_\phi$  [48], checked by enumeration at  $N_\phi = 64$  on the register's records. The entropic bound: the two Shannon entropies sum to at least  $\log_2 N_\phi$  [53], of the same form as the entropy identity of the ledger, bits read plus bits not read of  $u$  ( $H(u | K)$  for the outcome  $K$ ) =  $\log_2 N_\phi$  per record, the outcome's coarse graining ( $u$  stays in the world's list as a GameBoard diagnostic), read for the conjugate pair. The algebra: the Weyl relation of the shift and the phase multiplication, an identity of the group ring, the finite commutator in place of  $i\hbar$ . Kennard's variance form  $\Delta x \Delta p \geq \hbar/2$  [49, 52] follows by Cauchy–Schwarz on the Gram form of Theorem 4 in the limit  $N_\phi \rightarrow \infty$ , with  $\hbar = h/(2\pi) = h_q N_\phi/(2\pi)$ ; at finite  $N_\phi$  the bound is the finite group's. What makes it an uncertainty and not a spread is the one read-out: read at a Node a record gives its position and loses its phase, read on the fan's angle it gives its momentum's direction and not its Node, and the two are not jointly readable, as the law has no joint distribution for Bell (Theorem 6).

## 7 Bell and CHSH: from weights to counts

The correlation at the settings  $(a, b)$  is  $E = (c_{++} + c_{--} - c_{+-} - c_{-+})/N_\phi$  over  $N_\phi$  births, one per  $u$ , and the CHSH sum is  $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$  at the labels  $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$ . Two theorems follow from the definitions, exact on the GameBoard.

**Theorem 5** (Exact marginals; no-signalling). *For a pair with labels  $\{0, 1\}$  of equal weight and settings  $a, b$ , let  $J(o_A, o_B) = \sum_\ell U_a[o_A][\ell] U_b[o_B][\ell]$  and  $R(o_A, o_B) = J^2$ . Then the first party's outcome is  $+$  exactly for  $u < N_\phi/2$  and  $-$  for  $u \geq N_\phi/2$ , for every  $(a, b)$  and every  $N_\phi$ ; and the second party's count of  $+$  is exactly  $N_\phi/2$  for every  $(a, b)$  unless  $2N_\phi R(+, +)/C_K$  is an odd integer (a tie of the rung), in which case it is  $N_\phi/2 + 1$ . No tie occurs for any setting pair at the twelve grains from 8 to 1024 of [5], nor at the CHSH labels for any multiple of 8 up to 4096.*

*Proof.* Write  $n_a = C'[a]^2 + S'[a]^2$ . Since the rows of  $U_b$  are orthogonal on the integers,  $\sum_{o_B} J(o_A, o_B)^2 = n_a n_b$  for each  $o_A$ , so  $C_K = 2n_a n_b$  and the coarse cumulative after the first party's  $+$  is  $C_K/2$ :  $b = \lfloor (N_\phi C_K + C_K)/(2C_K) \rfloor = \lfloor (N_\phi + 1)/2 \rfloor = N_\phi/2$ . For the second party,  $R(o_A, o_B) = R(-o_A, -o_B)$  (the rows of  $U_s$  are each other's quarter turn), so with  $x = N_\phi R(+, +)/C_K$  the counts are  $c_{++} = \lfloor x + \frac{1}{2} \rfloor$  and  $c_{-+} = b_3 - N_\phi/2$  with  $b_3 = \lfloor (2N_\phi(C_K - R(+, +)) + C_K)/(2C_K) \rfloor = N_\phi + \lfloor \frac{1}{2} - x \rfloor$ . Hence  $c_{++} + c_{-+} = N_\phi/2 + \lfloor y \rfloor + \lfloor 1 - y \rfloor$  with  $y = x + \frac{1}{2}$ , which is  $N_\phi/2$  when  $y \notin \mathbb{Z}$  and  $N_\phi/2 + 1$  when  $y \in \mathbb{Z}$ . The tie condition is  $2x \in 2\mathbb{Z} + 1$ , that is  $2N_\phi R(+, +)/C_K$  odd. The absence of ties at the listed  $N_\phi$  is the computation over every setting pair [5].  $\square$

The strict-crossing rung  $N_\phi C_k > u C_K$  would give the second party 33/64 in 3944 of the 4096 setting pairs at  $N_\phi = 64$ , a signalling of  $1/N_\phi$  made by the rounding [3]; the nearest rung is what the theorem needs. The theorem's scope is what it states: a pair of two labels with equal weights, the CHSH form, and the rung of (9); unequal weights and more than two labels are not covered, and no-signalling for them is open. That the marginal at  $B$  is  $N_\phi/2$  for every  $a$  although  $B$ 's rung is computed from  $R(o_A, o_B)$  with  $a$  and  $o_A$  supplied by the layer (Definition 3) is the content of the theorem; the joint counts are not local, and the model does not claim they are.

**Theorem 6** (The finite- $N_\phi$  Bell value). *With the counts of Theorem 5 and off a tie,  $E_{N_\phi}(a, b) = 4c_{++}/N_\phi - 1$  and*

$$|E_{N_\phi}(a, b) - \cos(2\pi(a - b)/N_\phi)| \leq \frac{2}{N_\phi} + 4 \arcsin \frac{\sqrt{2}}{2\varrho} < \frac{2}{N_\phi} + 0.0111, \quad \varrho = N_t - \frac{\sqrt{2}}{2}, \quad (12)$$

*so at the labels  $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$ ,  $|S(N_\phi) - 2\sqrt{2}| \leq 8/N_\phi + 0.0444$ .  $S(N_\phi)$  is an exact rational for each  $N_\phi$ ; it lies above  $2\sqrt{2}$  for 252 of the 512 multiples of 8 up to 4096 (among them  $N_\phi = 16, 32$  with  $S = 3$  and  $N_\phi = 128$  with  $S = 23/8$ ) and equals  $181/64 = 2.828125$  at every power of two from 512 through 8192 and  $5793/2048 = 2.828613$  at 16384 [5]; measured after a detector at 512, 1024, 2048, 4096, 8192 and 16384 [4] (Figure 2).*

*Proof.*  $c_{++} = \lfloor x + \frac{1}{2} \rfloor$  with  $x = N_\phi R(+,+)/C_K$  gives  $|c_{++} - x| \leq \frac{1}{2}$  and  $E_{N_\phi} = 4c_{++}/N_\phi - 1$  from the mirror symmetry, so  $|E_{N_\phi} - (4R(+,+)/C_K - 1)| \leq 2/N_\phi$ . The vector  $(C'[s], S'[s])$  differs from  $256(\cos \theta_s, \sin \theta_s)$ ,  $\theta_s = \pi s/N_\phi$ , by at most  $\frac{1}{2}$  per component, so its direction differs from  $\theta_s$  by at most  $\delta = \arcsin(\sqrt{2}/(2\varrho))$  with  $\varrho = N_t - \sqrt{2}/2 = 255.29$ , below every table vector's length (each component within  $\frac{1}{2}$  of  $256 \cos$  or  $256 \sin$ ), so  $4\delta = 0.01108 < 0.0111$  and  $16\delta = 0.04432 < 0.0444$ ;  $J(+,+)/\sqrt{n_a n_b}$  is the cosine of the angle between the two directions,  $\cos(\theta_a - \theta_b + \epsilon)$  with  $|\epsilon| \leq 2\delta$ , and  $4R(+,+)/C_K - 1 = 2 \cos^2(\cdot) - 1 = \cos(2(\theta_a - \theta_b) + 2\epsilon)$ , within  $4\delta$  of  $\cos(2\pi(a - b)/N_\phi)$ . The four settings add the bounds. The values of  $S(N_\phi)$  are the computation from these formulas with the repository's tables [5], which reproduces the design's 176/64, 720/256 and 2896/1024 at  $N_\phi = 64, 256, 1024$  and the engine's three values below.  $\square$

What Theorem 6 does not say: that  $S(N_\phi) \leq 2\sqrt{2}$ , which is false for about half the  $N_\phi$ ; nor that the deficit is  $O(1/N_\phi)$  from below, which holds at the three  $N_\phi$  the engine ran and not in general. The tables' term in (12) does not shrink with  $N_\phi$ ; it is the resolution  $1/256$  of the circle's tables. At the CHSH labels the tables' entries are the same at every  $N_\phi$  (237, 98 and 181, 181), so  $E(0, N_\phi/8) = 46565/65773$  and  $E(N_\phi/4, N_\phi/8) = 46452/65773$  exactly at every  $N_\phi$  before the rounding of the cells; the rounding gives  $181/64$  at the powers of two from 512 through 8192 and  $5793/2048 = 2.828613$  at 16384 and 32768 (measured at 16384, Table 2), where the second pair of settings rounds to  $2893/4096$  and the tables' bound  $2N_\phi \leq 65536$  stops their computation ([5], `s_powers_of_two`) Beyond it the rungs on the fixed correlations are a closed form,  $S(N_\phi) = 8(c_1 + c'_1)/N_\phi - 4$  with  $c_1$  and  $c'_1$  the first cell's counts, the rungs of  $(1 + E)/4$  at the two pairs of settings [5]; the derivation's 24.4, the same values):  $11585/4096 = 2.828369$  at 65536 and  $370727/131072 = 2.828423$  at  $2^{20}$ , two-sided about the bound; with the tables fixed the limit in  $N_\phi$  is  $2(46565 + 46452)/65773 = 186034/65773 = 2.828425$ ,  $2.1 \times 10^{-6}$  below  $2\sqrt{2}$ , and  $2\sqrt{2}$  is the joint limit in  $N_\phi$  and  $N_t$  alone [5, 4]. At a tie  $c_{--} = c_{++} - 1$  and  $E_{N_\phi}$  drops by  $2/N_\phi$ ; no tie occurs at the CHSH labels up to 4096.

**The one prediction.** The law's number is

$$S_{U24} = \frac{181}{64} = 2.828125 \text{ exactly,} \quad \Delta S = S_{U24} - 2\sqrt{2} = -3.02 \times 10^{-4}, \quad (13)$$

at every power of two from 512 through 8192, with the tables at  $N_t = 256$ , the wheel at  $N_u = N_\phi$ , the settings  $(0, N_\phi/8, N_\phi/4, 3N_\phi/8)$  and a pair of two labels with equal weights; the marginals exactly  $1/2$ ; the correlations at the two pairs of settings  $725/1024$  and  $723/1024$  at  $N_\phi = 4096$ , above and below  $1/\sqrt{2}$ ;  $5793/2048$  at 16384 and 32768. It was computed in advance and measured after a detector at  $N_\phi = 512, 2048, 4096, 8192$  and, the plateau's end, 16384 (Table 2, the last three on main at `b34fe114`; Figure 2), under the click and the wheel with every pin met (the register's

block of the derivation’s 24.4); 512 and 4096 were re-run by the second runner [9]. The criterion: the law at a grain is rejected when a measured  $S$  differs from  $S(N_\phi)$  by more than five standard errors, under the identification of the experiment’s settings and detection with the labels and the ideal read-out, which the model does not model; the comparison with nature is Section 8.

## 8 The simulator’s checks and the comparison with nature

Three things are kept apart: the gates on the code; what is shown Inside by algebra under named assumptions, which needs no experiment; and what is measured, a run Inside read Outside, compared with known experiments.

**The code implements the law.** Every registered world is run from its world file with its expectation written before the run and its source fingerprint; a second runner re-ran series L, T and fifteen more worlds bit by bit [9]; the books balance at every interval in every run; the rounding pins of the (3, 4) split were met (Table 2); the crossing rule’s tests read their pinned counts. These are gates on the code, not measurements.

**The formulas after a detector.** Every number of Table 2 is a count of clicks or a click’s tick at a declared detector, registered with its fingerprint; a number of the GameBoard’s state appears nowhere. Seven formulas shown Inside (the pace, the click’s power, Mach-Zehnder, the marginals, GHZ, Malus, light beside a mass) were confirmed by their runs, each number pinned before the run, and are listed in Appendix C with their series and fingerprints; Table 2 keeps the readings that no formula gives.

Table 2: The representative detector checks of the formulas. Series L’s runs carry the source fingerprint `ff5c382d672f`, L7’s `4bf55a62e6fd`, series S’s `24e0c1ba`, series T’s `afb533a` [4, 9]; each expectation was written before its run.

| Formula                                                                      | Run (series, world)                                                                                                                                                      | Detector reading                                                                                                                                                                                                                                                                                                                           | Expected before the run                                                              |
|------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Young’s spacing                                                              | L2b, two slits under the birth wheel, 4096 births                                                                                                                        | the bright pixels 19 to 51, the dark 0 to 3, the visibility 0.966 in the clicks                                                                                                                                                                                                                                                            | the dark pixels as pinned; the rungs within one                                      |
| $S(N_\phi)$                                                                  | L and L6, the pair at $N_\phi = 64, 512, 1024, 2048, 4096, 8192, 16384$ (the last three on main at <code>b34fe114</code> )                                               | 176/64; 1448/512; 2896/1024; 5792/2048; 11584/4096; 23168/8192; 46344/16384                                                                                                                                                                                                                                                                | the same, Theorem 6 and its closed form: the plateau 181/64 and its end 5793/2048    |
| The clock’s field at two Nodes (the continuum’s $1/r$ its limit)             | T, a lamp’s births at a detector, two sources at 3 and at 6 Links                                                                                                        | the ratio of the two shifts 1.907 (age word); 1.000 (presence word)                                                                                                                                                                                                                                                                        | the lines’ pin $1.909 \pm 0.05$ ; replicated                                         |
| The equivalence principle after a detector; the $1/r$ force’s scale symmetry | D3, a probe carrying a lamp on series D’s plane at $r = 12$ and 24, its births read at a line of one-Node detectors; the held mass four times at $r = 24$ ; two controls | the same $x$ on 138 of 139 common birth ticks (one Node the largest difference) and the same escape tick at four times the mass; $T(24)/T(12) = 1.997$ from one recurrence per radius on loops that are not similar figures; the controls at $x = 60 + r$ on every click; the circle’s period, amplitude and $\omega^2$ outside their pins | one birth interval and one Node; $2.00 \pm 0.18$ ; $60 + r$ , the escape $278 \pm 6$ |

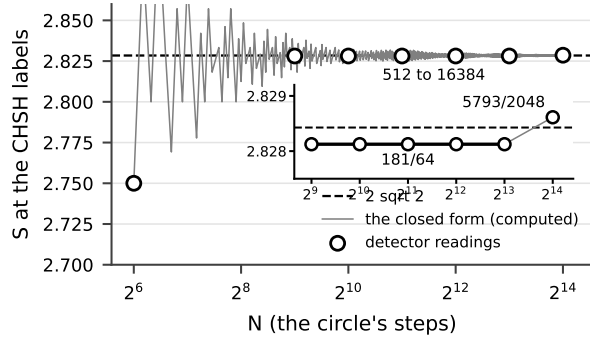


Figure 2:  $S(N_\phi)$  at the CHSH labels. The curve is the closed form  $S(N_\phi) = 8(c_1 + c'_1)/N_\phi - 4$  computed from the rule at every multiple of 8 with the tables at  $N_t = 256$ , a computation [5]; the circles are detector readings (DETECTOR) at  $N_\phi = 64, 512, 1024, 2048, 4096, 8192$  and  $16384$  (series L, L6 and the register's block of the derivation's 24.4, the last three on main at **b34fe114**); the inset marks the plateau  $181/64$  from  $512$  through  $8192$  and its end  $5793/2048$  at  $16384$ ; the dashed line is  $2\sqrt{2}$ .

**The comparison with nature.** The confrontation register [26] keeps one row per registered detector reading with a dimensionless counterpart in nature, under one parameter set, one published source per value and one verdict per row: PASS within the stated uncertainty, FAIL with the number, BOUND where the comparison bounds a free parameter, NOT COMPARED where the law does not compute the observable. A FAIL is a result of the law and is stated with the same care as a pass. Table 3 carries the rows read after a detector; seven rows of the register (4a, 5b, 6, 10, 11a to 11c) rest on a pin whose run is not made and carry no number here. Every verdict is the law's as built at the merge commit named in Appendix C; a hypothesis's PASS beside the law never changes the law's FAIL.

Table 3: The confrontation register's rows read after a detector [26]. Three PASS (replicated [9]), a fourth under the clock's assumed word (12); twelve FAIL (seven in a registered run, four by a pin without its run and not in this table, one a declared input refuted, 8c); two BOUND (5a, 7a); one NOT COMPARED (2c).

| Row | Observable and nature's value                                                                                                                                                 | The law's detector reading                                                                              | Verdict                                                                                                                                                                                          |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1a  | The CHSH sum of a pair: $2.42 \pm 0.20$ [65]; the quantum bound $2.828$                                                                                                       | $S = 2.75$ at $N_\phi = 64$ , series L                                                                  | PASS: 1.65 standard errors above the measured, 0.078 below the bound                                                                                                                             |
| 1b  | The same, the phase-form window                                                                                                                                               | $S = 2$ exactly                                                                                         | FAIL: the local bound, 2.1 standard errors below                                                                                                                                                 |
| 2a  | Two-slit visibility of one quantum at a time: 0.98 [66], the ideal 1; the criterion the clicks' visibility $(I_{\max} - I_{\min})/(I_{\max} + I_{\min})$ , no apparatus model | 0.966 in the clicks of slits_huygens (the birth wheel, 4096 births; L2b), 0.954 in the record's weights | FAIL: 0.014 below the measured, 0.034 below the ideal; the cause named, the screen's fan's grain (the pin derived for the screen's fan, not this world's)                                        |
| 2b  | Mach-Zehnder visibility: 0.98 [66]; the criterion the visibility $(b - d)/(b + d)$ of the bright and dark ports' offers of <b>mz.equal</b> , no apparatus model               | the offers 1681/1682 and 1/1682; the clicks 64/0 over 64 births                                         | PASS: 0.9988, above 0.98                                                                                                                                                                         |
| 2c  | The power of the click's form: the Sorkin parameter $\kappa = 0.0064 \pm 0.0119$ [39]                                                                                         | the power's window [1.917, 2.012] from the click cells, a read-back of the built click                  | NOT COMPARED ( $\kappa$ not computed): the window an implementation gate, no evidence about nature; no three-opening world is registered                                                         |
| 3   | The deceleration parameter $q_0 = -0.53 \pm 0.01$ [40, 41, 42]                                                                                                                | $-0.108$ coasting (series G2, the pointer's $z$ ; $-0.104$ at head, record 408 [6])                     | FAIL: no term with $q < 0$                                                                                                                                                                       |
| 4b  | A moving lamp's redshift with the clock's factor [36]                                                                                                                         | $z = 0.2636$ at $\beta = v/c = 0.2674$ (series G2; the re-run at head pending)                          | FAIL: 0.315 expected, 17 grains below; under covariant-readings-v1 0.3674 for the pinned $0.369 \pm 0.003$ (series S), the law's row FAIL; the frame: the two one-way factors apart by $1 - v^2$ |

| Row | Observable and nature's value                                                                                                           | The law's detector reading                                                                                                               | Verdict                                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5a  | The anisotropy of $c$ by direction, below $10^{-18}$ [35]                                                                               | the flight table's pace 0.5774 to 0.5818 at $N_t = 64$ (series Q within it at finite ages, Appendix C)                                   | BOUND on $N_t$ : of order $5.8 \times 10^{17}$                                                                                                        |
| 7a  | The deuteron's binding fraction 0.1185 percent [37]                                                                                     | the escaped content and the mass a detector reads on the lifetime border (series N): the held bond 0.109 percent                         | BOUND on a declared input                                                                                                                             |
| 7b  | The alpha's binding over the deuteron's, 12.72 [37]                                                                                     | 2.0 (series N, the border's clicks)                                                                                                      | FAIL: a factor 6.4; the give once per body                                                                                                            |
| 8a  | The neutron's decay curve, exponential, width over median 3.17 [45]                                                                     | 0.036, a step (series J1, 64 clicks)                                                                                                     | FAIL: a factor 88                                                                                                                                     |
| 8b  | The neutrino's passage through a second detector, about 1                                                                               | 16 of 1024 with 0 behind (series J2, a window)                                                                                           | FAIL                                                                                                                                                  |
| 8c  | The heaviest neutrino mass state over the electron's mass: at least $9.8 \times 10^{-8}$ [43, 44]                                       | the nu family with no content, an input                                                                                                  | a declared input refuted by nature: massless states cannot carry the observed splittings; a content above 0 is an input the register does not declare |
| 9   | Malus's law at 45 degrees, 1/2                                                                                                          | 128 of 256 exactly; 219 of 256 at 22.5 degrees (A12)                                                                                     | PASS: exact at 45 and 90 degrees; 0.8555 against $\cos^2 = 0.8536$ at 22.5                                                                            |
| 12  | The clock's field at two distances at the same push: the potential's form, 2.00 (the GPS term 45.7 microseconds per day [14]; [15, 38]) | series T: the form measured at a detector at two Nodes, the age clock 1.907 (the lines' pin $1.909 \pm 0.05$ ), the presence clock 1.000 | PASS under the age word (P9), nature's 2.00 4.6 percent away; FAIL under the presence word                                                            |

**The prediction, and the failures.** Against the photon-pair value  $S = 2.82759 \pm 0.00051$  [64], under three assumptions the paper states and does not defend (nature's pairs this model's pairs at some  $N_\phi$ , fair sampling, the four settings the model's labels), the law's 181/64 stands 1.05 standard errors above, 16384 and beyond at 1.5 to 2.0;  $N_\phi = 64$  and 256 are excluded; the loophole-free [65] excludes no  $N_\phi$ . The proximity is a compatibility, not an advantage over quantum mechanics; what refutes it is a CHSH measurement with an uncertainty below  $1 \times 10^{-4}$  reading a deficit below  $2 \times 10^{-4}$  or above  $4 \times 10^{-4}$ . The failures are the law's own results (Table 3; light unbent is Section 5's, against 1.75 arcseconds): the unslowed clock, the deceleration, light unbent, the weak forms, the strong ratio, the two-slit visibility, the phase-form window, the massless neutrino; each refutes the law as declared on its own, and no count of passes weighs against them.

## 9 Discussion and conclusions

**What is proved.** Derived Inside, exact on the GameBoard, no experiment needed: the group of the six Ports and its 24 rotations; the books' conservation and the lattice form of the continuity equation; Gauss's law of a free family's flux under its world condition; the split's isometry and the injectivity of the interval on the amplitudes for a fixed event history; the pace of every direction, isotropic within  $1/T_D$ ; the click's weight a positive quadratic form of power 2; the exact marginals of a pair; the CHSH sum as an exact function of the grain and the tables; Planck's and de Broglie's relations as identities under the declared dictionary. Recovered in the limit of the grain, with a rate: Born's rule to  $1/N_\phi$  per cell and the cumulative rungs to  $1/(2N_\phi)$ , Tsirelson's bound to  $8/N_\phi$  plus the tables' 0.0444 ( $2\sqrt{2}$  in the joint limit of  $N_\phi$  and  $N_t$ ), Young's spacing, the attained value of  $c$ , the continuity equation's differential form. Recovered under an average that covers the shell, with the condition in the claim: Newton's and Coulomb's inverse square, the retarded potential and Poisson's equation, the clock's  $1/r$  form. Each names the known form the algebra recovers (Gauss's, Newton's, Coulomb's, Born's, Tsirelson's, Planck's, Einstein's); the paper's own are the integer road, the finite- $N_\phi$  values (Born's bound  $1/N_\phi$ , the plateau 181/64, the fixed-table limit),

the lattice Gleason on  $\mathbb{Z}_{N_\phi}$ , the measurable  $1 - v^2$  and the entropy identity of the click; the value  $1/\sqrt{3}$ , Tsirelson's limit and the dictionary's identities claim no novelty.

**What the runs support and what nature says.** After a detector the simulator reads Table 2's list, each inside an expectation written before the run; against nature three readings pass, a fourth under the clock's assumed word, twelve fail and one is not compared (Table 3).

**The distance, stated.** What the broad claim still needs, and the check that decides each:

Table 4: What no rule of the law forces, and the check that decides each.

| Needed                                                                                | The deciding check                                                                                                                                                                   |
|---------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Newton's inverse square and Poisson's law closed after a detector ( $G$ not read; D3) | a closed orbit's period at two radii under the directional drive (form B, not built); the escape's count at a face detector; the clock's form at more distances                      |
| The Lorentz factor: the law meets (A1) and not (A2)                                   | the ratio of the two one-way Doppler factors through records: $1 - v^2$ under the law, 1 under Lorentz; the split of a row between staying and hopping (A2), a hypothesis not opened |
| The bending of light: the flight is blind to the crowd                                | a rule under which a row's wall reads the crowd's age moment (a hypothesis, its coefficient an input) against nature's 1.75 arcseconds                                               |
| The masses' values: every rule linear in the content, every equal split a fixed point | a nonlinear closure on the amounts under its own identity; until then the family table is an input                                                                                   |
| The click's harmonic constants; the circle as position (a hypothesis, Section 6)      | the least-rank click, an axiom of the apparatus pinned by the two-slit period and Malus at 22.5 degrees; the single opening against its pin (row 10)                                 |
| A CHSH measurement at $10^{-4}$                                                       | decides 181/64 against $2\sqrt{2}$                                                                                                                                                   |

**Inside and Outside, and how a click arrives at Lorentz.** A packet moved Inside, from place to place, and what it says Outside: Inside, the law propagates records as a beam by the algebraic formulas of Section 2, where no one measures; Outside, the game above the board, there are detectors and their clicks only, the beam passing to them with amplitudes at the click (P11). A detector's clock is what it emits and receives back, on its own record; no Node holds a detector's time; an open face, without a body, has no clock, its tick the record's ordering (a GameBoard diagnostic); a velocity is Nodes apart over counts apart between neighbouring detectors, and the read-out of Definition 3 is the click. The frame [25]: "To convert the lattice and the momentum inside it into motion Outside, the game above the board, one uses clicks that pass information, a click being the passage of information from Node to Node at most one Node per interval (A1) and its content an amplitude with a phase that splits each interval between staying and hopping, the mass the staying share (A2); Lorentz of the world is assumed, and what is shown is that this conversion brings it. The symmetry of the conversion, the transformations between click families that preserve (A1) and (A2), is the Lorentz group up to scale, acting on the amplitudes of the passing packet as the Dirac walk's covariance, so that the passage of clicks is Lorentz up to corrections of order  $m^2 v^2$ , exact in the continuum limit, a detector's own rate  $\sqrt{1 - v^2} (1 - \kappa^2/6)$  with the mass angle  $m$  the spacing over the reduced Compton wavelength; in three dimensions the lattice's anisotropy begins at the fourth order. The conversion is made exactly with the amplitudes, converting the packet that passes, and it comes out from there; the law as built satisfies (A1) and not (A2)." (A1) and the one line, the relativity of the two directions ( $k_{AB} = k_{BA}$ ), are special relativity's two postulates in the click language, so what is arrived at is Einstein's road walked on clicks, and what

the board adds is (A2), the amplitudes that carry  $\sqrt{1-v^2}$  into a detector's own rate. The law as built meets (A1), one Node per interval with the flight blind to the phase (P9), and not (A2), its hop a whole-record schedule with no staying amplitude; so its clicks carry the group and its counter runs at the rest rate at every speed (rows 4a, 4b): read through records, the two one-way Doppler factors  $r/(1-v)$  and  $(1+v)/r$  are alike only at  $r^2 = 1-v^2$ , and the law's  $r = 1$  leaves them apart by  $1-v^2$ , the measurable that decides. No wall of the law and no declared identity enter the reading; Eq. (14) is an Inside formula on the record, read by no detector, whose Outside formulas the conversion gives. What the system gives is a computation Inside whose passage Outside is the clicks, at the quantum level as well; how far Outside represents reality is what Tables 2 and 3 show, result by result.

**The step beneath and the step above.** The paper's centre is the step beneath the lattice, Eq. (14) in its frame, the six operations on the record with  $W$  its invariant; the direction is one way: the step beneath is the axiom, the conversion of (A1) and (A2) is the map, and the step above is its image, a theorem named as what Einstein wrote. From  $W$ , by the conversion, the step above follows:  $Wc^4$  is  $E^2 = E_0^2 + p^2c^2$ , the gate  $E_0/E'$  is  $1/\gamma$ , a detector's own rate, the drive's fraction is the click's velocity, and the two Doppler factors are the step above read at two detectors. The click frame's section 7 [25] shows the chain with no square declared: the walk's exact invariant  $\cos \omega = \cos m \cos \kappa$  gives  $\omega^2 = m^2 + \kappa^2 - m^2\kappa^2/3 + O(6)$ ; the law's own Planck map ( $E = h_q n/d = h_A f$ ,  $p = \hbar \kappa$  per Link,  $E_0 = \hbar m$ , the load identity  $3\hbar n = Q S d$ ) gives  $E^2 = E_0^2 + c^2 p^2$ ; the whole unit  $E' = E/c^2$  gives  $W = E_0'^2 + 3\mathbf{p} \cdot \mathbf{p}$  with  $3 = 1/c^2 = d$  exactly on the body diagonal (2.954, 2.971, 3.000 by direction, the root's rounding), the corrections of relative order  $m^2\beta^2$ , exact in the continuum limit. So  $W$  is shown to that order as the conversion's identity under (A1) and (A2), no square declared anywhere in the chain [23], and Einstein's relation follows from the Inside alone; for the rows' dynamics, which hop whole and lack (A2),  $W$  remains the declared identity of record 270 [6], the reading's identity and not yet the rows'. The same road serves Newton (below) and, in the same font, what it does not reach: length contraction, the perihelion's second order, the bending of light, waves, horizons and the field equations [21].

**The smallest thing above.** What the Outside step is made of follows from (A1) and the click's definition with no run: a click is a Node, a detector's own count and what arrived, and the counts of a family add, so their ratios lie in  $\mathbb{Q}$  [25]. Hence the least distance a click can report is one Link, Nodes apart being an integer; the least time a detector times is one count, a pulse and its return two; the least step of a velocity at  $k$  counts is one Node per  $k$  counts;  $c$  is one Link per the least count; and the amount  $w$  of a row is the unit that arrives. Every reading Outside is a whole number of these, which is why the world above the board is quantized: physics above cannot take this from its own formulas, and the Inside gives it. The speed of light Outside is then the arrival count over the Euclidean distance,  $c_D = N_l|D|/T_D$  by a row's age or by a pulse and its return, its anisotropy  $1/c_D^2 = 2.954, 2.971, 3.000$  on the heading, the face diagonal and the body diagonal and  $1/\sqrt{3}$  in the limit [20].

**Newton from the small step.** The same way, with five assumptions on main today and no (A2) (the crowd read as the age moment, the age wall at coefficient 1, the push, the flight blind, the reading a pulse and its return) and the spreading of  $K$  beams over a shell of  $N(r)$  Nodes, which is arithmetic and not a law, the fall, the equivalence principle, Kepler's ratio and the clock in a crowd come out Outside, the inverse square at the shell mean's rung and the equivalence exactly,  $M_A$  cancelling record by record; the register's D3, T and X are their confirmation, and the inverse square's decisive reading is not made [25]. Every result Outside is thus the image of one Inside

formula under the conversion, and Table 5 lists the map row by row with the reading that sits on the Outside side.

Table 5: Inside formulas and Outside formulas, the conversion between them with its order, and the register's reading on the Outside side (the click frame's section 7 [25]; SHOWN is algebra under the named assumptions, MEASURED a run Inside read Outside).

| Inside (from the six operations)                                                                                                      | Outside (a detector reading)                                                                                    | The conversion; its order                                                         | The reading (kind)                                                                                  |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| $W = E_0'^2 + 3 \mathbf{p} \cdot \mathbf{p}$ on the record, $E'$ the whole root by comparisons; on the law no square, the wall linear | the energy a click reads, $\gamma$ , the rate $r = E_0'/E'$ (the identity) or 1 (the law)                       | (A1), (A2), the Planck map; shown to order $m^2\beta^2$ ; the law's $r = 1$ exact | series S: the 64th self-creation at 70 and 124 (GAMEBOARD); $r$ not read                            |
| the pace $ p_a /(N_l N_w M +  p_a )$ per axis; under the identity $p/E'$                                                              | the velocity: Nodes apart over counts apart between neighbouring detectors                                      | (A1); exact within $1/T_D$                                                        | D3's controls at their pace on every click (DETECTOR)                                               |
| the phase per age $n/d$ , $E = h_q n/d = h_A f$ , $\lambda = h_A/p$                                                                   | the frequency a detector counts, $1+z$ ; $k_{BA} = (1+v)/r$ , $k_{AB} = r/(1-v)$ , the round trip $(1+v)/(1-v)$ | (A1) and the count, rung 1                                                        | row 4b 0.2636 (FAIL); series S 0.3674 (MET in its domain); $k_{AB}$ not read                        |
| the age wall, the crowd $a_\tau$ at coefficient 1, the rate $1/(1 + a_\tau n/d)$                                                      | the clock rate in a crowd by a pulse and its return, the ratio of two lamps' $1+z$                              | (A1) and the count, rung 1                                                        | series T 1.907 for 1.909 $\pm$ 0.05; series X $k = 0.9089$ at $r = 4$ for the pin 0.9108 (DETECTOR) |
| the flight table $T_D$ , the pace $N_l D /T_D$                                                                                        | the arrival count at a detector $L$ Links away; $c$ one Link per the least count                                | (A1); rung 1 in the count, rung 2 in the isotropy                                 | series Q 290 of 290; series S 369, 345 (DETECTOR)                                                   |
| the click's bilinear form $\mathbf{f}^T \mathbf{G} \mathbf{f}$                                                                        | Born's form, the count of clicks over many records                                                              | (A2) at the click; rung 1                                                         | the two-slit fringes, Malus (DETECTOR); the amount and the phase of one record never pass singly    |
| the spreading, the presence $q\tau_L/(4\pi r^2)$ and the age moment $q\tau_L/(4\pi cr)$ in the shell mean                             | the clock's field, the ratio of two shifts 2.00; a shell's flat interior                                        | rung 2, the shell mean; the ripple MEASURED ONLY                                  | series T 1.907; series X 1.0029 for 1.0039 (DETECTOR)                                               |
| the push $-M_A \mathbf{a}$ and the drive                                                                                              | the fall, $a = -GM_B/r^2$ , $G = K\eta/(4\pi N_w)$ ; the equivalence                                            | (A1) and the count; the equivalence rung 1, the inverse square rung 2             | D3: 138 of 139 common birth ticks (DETECTOR); the inverse square's reading not made                 |
| the circular momentum under the push, $v^2/r = GM_B/r^2$                                                                              | Kepler's period, $T(24)/T(12) = 2$ on the plane                                                                 | rung 2, $v \ll c$                                                                 | D3: 1.997 in [1.82, 2.18] (DETECTOR)                                                                |

**The road to each, and the check that nothing was derived from them.** Three roads are shown, each from the Inside step, the six verbs, (A1) and, where it enters, (A2), with (A3), locality Outside (nothing Outside passes faster than a chain of clicks or jumps a Node), a definition of Outside and not a law [21]. To Lorentz: two legs at one Node per interval, the two directions alike ( $k_{AB} = k_{BA}$ ), give the boost with  $v = (k^2 - 1)/(k^2 + 1)$ , the scale  $r$  left free, the click frame's section 2; (A2) is what fixes  $r$  at  $\sqrt{1 - v^2}$ . To Einstein: the place-to-place factor  $k_{XY}$  from (A1) and the counts, the radar map, the rate  $1/\gamma$  as  $r$  read from two factors, the composition of velocities  $r$ -free, and  $E^2 = E_0^2 + c^2 p^2$  as the line of the rate with  $W$  shown to second order (Section 7 of the frame), Theorems 1 and 2 and rows II.1, II.3 and II.6 of [21]. To the step above itself: the Outside step, place apart and count apart of a detector between two clicks, is the image of the Inside step under the conversion, Einstein's step its most general form where  $r$  is supplied, its quantum a theorem of (A1), (A3) and the drive's one Link per self-creation, and Newton's step its limit at  $v \ll c$  (section 3 of [21]). Einstein's, Lorentz's and the Outside step's formulas appear in these chains only as the thing compared with; no chain takes them as an input, and the Outside step is never a second formula set beside the Inside step. The word is "follows" or "recovered" where a known

form is arrived at, “shown to second order” for  $W$  under (A1) and (A2), “declared” for  $W$  on the rows’ dynamics and for the unit  $E_0 = mc^2$ ; “derived” stands nowhere in the three roads. Table 8 (Appendix D) lists the check row by row: the inputs each chain used, the file and line where it starts, and the word; the check was made by reading each chain against the certification tables of the two sources, and the physics-rule reviewer audits the same in the sources.

**What the Inside step gives Outside that the continuum cannot state.** Formulas of the six verbs taken Outside by the conversion, each with the click that reads it and its kind; none of them is a formula of the continuum. (i) The quantum of the Outside step: one Link, a pulse and its return, a velocity  $1/k$  in the mean with its quantum  $1/(k(k+1))$  at the pace  $1/k$  (Theorem 3 of [21]); read as a least step in series S’s face clicks 369 and 345 within two of the flight table’s and D3’s 138 of 139 births at the same Node (DETECTOR); the velocity’s quantum and the pulse’s two counts not read as such. (ii) The pair’s sum at the CHSH labels is the exact rational the rung gives,  $|S(N_\phi) - 2\sqrt{2}| \leq 8/N_\phi + 0.0444$ , above the bound at  $N_\phi = 16$  and  $32$  ( $S = 3$ ), the plateau  $181/64$  from 512 through 8192 below it by  $3.02 \times 10^{-4}$  and  $5793/2048$  above at 16384 (Theorem 6; the frame’s section 10); measured at seven  $N_\phi$  (Table 2, DETECTOR). (iii) The perihelion: the drive’s pace departs from  $p/m_i$  at first order per axis, so a bound orbit’s apsides move per revolution by an angle of order  $\beta$  with the lattice’s symmetry, where Einstein has  $6\pi\beta^2$  isotropic, larger than nature’s by  $1/(6\pi\beta)$ , about 330 times at Mercury’s pace if its coefficient is of order one, and of the wrong symmetry: FAIL on the law, pinned in order and symmetry, the coefficient NOT MADE; under the identity  $\pi\beta^2$ , one sixth; the deciding reading, the apsides of a lamp’s orbit at two orientations to the axes, NOT MADE, its pin written before any run, an apsidal motion of order  $\beta$  that changes with the orientation and not with the mass held ([21], II.12). (iv) The grain: the radial residence factor times the pace,  $\tau_L c = 0.993$  on a heading, multiplies every constant read at a finite grain and is 1 in the limit of every direction. (v) The anisotropy of  $c$  by direction,  $1/c_D^2 = 2.954, 2.971, 3.000$  (series Q, 290 of 290, DETECTOR; [20]). (vi) The equivalence’s constant: the clock in a crowd and the accelerated detector read one shift,  $\delta k = (nS/d)(gY/c^2)$ , Einstein’s  $gY/c^2$  exactly when  $nS = d$ , a condition on an Inside declaration and not a result; the register’s clock and crowd worlds declare  $nS/d = 16$  (GAMEBOARD, the world files), so the condition is not met by them, the form’s readings are series T’s 1.907 and series X’s  $k$  (DETECTOR), and the constant’s reading is NOT MADE ([21], II.10). (vii) Light’s bending: on the law 0, the flight blind to the crowd (series K, 0.000 pixel, DETECTOR; FAIL against 1.75 arcseconds); under the key `optical`, the flight in the age wall’s set with a declared  $c_f$ , the delay  $c_f(nS/d)(GM/c^3)\ln(4r_1r_2/b^2)$  and the bending  $\alpha = 2c_f(nS/d)GM/(c^2b)$ , Einstein’s at  $nS = d$  and  $c_f = 2$ , NOT COMPARED, the route the owner’s decision ([21], II.11). The families of the register, each with its declared integers, its algebraic object and the click that reads it, are Table 7.

**The limits, and what does not return.** The content and  $\sum w^2/\mathfrak{m}$  are conserved exactly, the offered norm only within the tables’ rounding. The covariant readings (record 270 [6]) read one Inside formula Outside on one axis, series S against its pins: a body’s record carries the exact square of its energy,

$$\boxed{W = E_0^2 + 3\mathbf{p} \cdot \mathbf{p}, \quad E_0 = N_l N_w M, \quad c^2 = 1/3} \quad (14)$$

compared and never rooted, with  $E'$  the largest integer whose square is at most  $W$  and every count of the body gated by  $E_0/E'$ ; multiplied by  $c^4$  it is the energy–momentum relation of special relativity (series S, Table 5, inside their pins). The hypotheses beside the law are on the tree [11].

**Positioning.** The click is a non-local step, forced by Bell's theorem because the marginals are exact and  $S > 2$  [27, 28]; no-signalling is Theorem 5 [29]. The click's finite kinematics is Weyl's and Schwinger's on  $\mathbb{Z}_{N_\phi}$  [50, 51]; the transport a lattice gas's exact integer dynamics [30, 31], free of dispersion where the lattice automata of [55, 56, 57] disperse; the uniform  $u$  does the work of quantum equilibrium [58, 59]; the model is  $\psi$ -ontic [60, 61]; Theorem 6 puts  $S(N_\phi)$  on both sides of Tsirelson's bound [62, 63]. The earlier testbed [1] is history.

## References

- [1] A. Gonen, A finite-integer causal-GameBoard testbed: which resources reproduce interference, causal collapse and Bell correlations on one GameBoard (2026), arXiv preprint.
- [2] The Beam Law and its implementation notes, `docs/BEAM_LAW.md` of the archived code [67].
- [3] The design of `amplitude-v1` and its check outputs, `docs/designs/amplitude-v1/` of the archived code [67].
- [4] The experiments register, `docs/EXPERIMENTS.md` of the archived code [67].
- [5] `paper/general_formula/checks/s_of_n.py`, `s_powers_of_two.py` (the pair's  $S$  and its four correlations at the powers of two from 512 to 32768, the closed form on the fixed correlations to  $2^{20}$  with the distances from [64], with the tables fixed), `which_path_window.py`, `tables_norm.py`, `information_transfer.py`, `limits.py` and `light_speed.py` with their outputs, `figures/summary.json` and `NUMBERS.md`, in the archived code [67].
- [6] The dated log of 2026-09-20, `docs/LOG_2026-09-20.md` of the archived code [67].
- [7] C. E. Shannon, Bell Syst. Tech. J. 27, 379 and 623 (1948).
- [8] The one symbol table of the law without clashes with physics' letters, the derivation mathematician's proposal `docs/designs/notation/NOTATION.md` of the archived code [67].
- [9] The replications register, `docs/REPLICATIONS.md` of the archived code [67].
- [10] The families in the paper, the mathematician's audit of 2026-09-21, `docs/designs/paper_families/FAMILY_TABLE_AUDIT.md` with its host script's output `family_table.out`, of the archived code [67] at the commit named in Reproducibility.
- [11] Hypotheses under test, entry 27, the conditional derivations as declared hypotheses, `docs/HYPOTHESES.md` of the archived code [67].
- [12] Canonical simulation terminology, `docs/TERMINOLOGY.md` of the archived code [67].
- [13] The clock's word, the presence or the age moment: the physicist's read, `docs/designs/clock_age/NOTE.md` of the archived code [67].
- [14] N. Ashby, Living Rev. Relativity 6, 1 (2003).
- [15] R. V. Pound and G. A. Rebka, Phys. Rev. Lett. 4, 337 (1960).
- [16] The law as one vector algebra: every rule in force as a vector operation, with its integer form beside it, `docs/designs/vector_form/LAW.md` of the archived code [67].
- [17] The engine's document, `docs/ENGINE.md` of the archived code [67].
- [18] The project's decisions and dictionary, `docs/HIGHLIGHTS.md` of the archived code [67].
- [19] The derivations of the beam law, `docs/DERIVATIONS_BEAM.md` of the archived code [67].
- [20] Light Outside from the board: every equation of light Inside carried Outside by its own transformation, `docs/designs/light_outside/DERIVATION.md` of the archived code [67], at ceb6e066 (PR #791, pending merge).
- [21] Einstein Outside: the formulas of the special and the general theory brought from inside the GameBoard to above it, `docs/designs/einstein_outside/DERIVATION.md` of the archived code [67], on main at e8432e6c (PR #792, merged).
- [22] The transition from ordinary physics to modern algebra: a dated, cited history, `docs/designs/algebra_transition/HISTORY.md` of the archived code [67], on main at 412f5c61 (PR #793, merged).

- [23] The click frame, part 2 (“no square is declared anywhere in the chain”), `docs/designs/click_frame/DERIVATION.md` of the archived code [67] on `main` at `af051aaf` (PR #801, merged), one file with the click frame.
- [24] T. Toffoli and N. Margolus, *Cellular Automata Machines* (MIT Press, Cambridge, 1987).
- [25] Lorentz from the clicks: the click frame, its section 0 the click theorem, `docs/designs/click_frame/DERIVATION.md` of the archived code [67], on `main` at `70e9781a` (PR #769, merged; the file identical to its head `1dd81fef`).
- [26] The confrontation register, `docs/NATURE.md` of the archived code [67].
- [27] J. S. Bell, *Physics* 1, 195 (1964).
- [28] J. F. Clauser, M. A. Horne, A. Shimony and R. A. Holt, *Phys. Rev. Lett.* 23, 880 (1969).
- [29] G. C. Ghirardi, A. Rimini and T. Weber, *Lett. Nuovo Cimento* 27, 293 (1980).
- [30] J. Hardy, O. de Pazzis and Y. Pomeau, *Phys. Rev. A* 13, 1949 (1976).
- [31] U. Frisch, B. Hasslacher and Y. Pomeau, *Phys. Rev. Lett.* 56, 1505 (1986).
- [32] R. Courant, K. Friedrichs and H. Lewy, *Math. Ann.* 100, 32 (1928).
- [33] Y. H. Qian, D. d’Humières and P. Lallemand, *Europhys. Lett.* 17, 479 (1992).
- [34] A. A. Abdo et al. (Fermi LAT and Fermi GBM Collaborations), *Nature* 462, 331 (2009).
- [35] M. Nagel, S. R. Parker, E. V. Kovalchuk, P. L. Stanwix, J. G. Hartnett, E. N. Ivanov, A. Peters and M. E. Tobar, *Nature Commun.* 6, 8174 (2015).
- [36] B. Botermann et al., *Phys. Rev. Lett.* 113, 120405 (2014).
- [37] M. Wang, W. J. Huang, F. G. Kondev, G. Audi and S. Naimi, *Chin. Phys. C* 45, 030003 (2021).
- [38] P. Delva et al., *Phys. Rev. Lett.* 121, 231101 (2018).
- [39] U. Sinha, C. Couteau, T. Jennewein, R. Laflamme and G. Weihs, *Science* 329, 418 (2010).
- [40] N. Aghanim et al. (Planck Collaboration), *Astron. Astrophys.* 641, A6 (2020).
- [41] A. G. Riess et al., *Astron. J.* 116, 1009 (1998).
- [42] S. Perlmutter et al., *Astrophys. J.* 517, 565 (1999).
- [43] S. Navas et al. (Particle Data Group), *Phys. Rev. D* 110, 030001 (2024).
- [44] M. Aker et al. (KATRIN Collaboration), *Nature Phys.* 18, 160 (2022).
- [45] F. M. Gonzalez et al., *Phys. Rev. Lett.* 127, 162501 (2021).
- [46] M. N. Huxley, *Proc. London Math. Soc.* 87, 591 (2003).
- [47] A. M. Gleason, *J. Math. Mech.* 6, 885 (1957).
- [48] D. L. Donoho and P. B. Stark, *SIAM J. Appl. Math.* 49, 906 (1989).
- [49] E. H. Kennard, *Z. Phys.* 44, 326 (1927).
- [50] H. Weyl, *The Theory of Groups and Quantum Mechanics*, translated by H. P. Robertson (Methuen, London, 1931; Dover, New York, 1950), chapter IV, section 14.
- [51] J. Schwinger, *Proc. Natl. Acad. Sci. USA* 46, 570 (1960).
- [52] H. P. Robertson, *Phys. Rev.* 34, 163 (1929).
- [53] H. Maassen and J. B. M. Uffink, *Phys. Rev. Lett.* 60, 1103 (1988).
- [54] P. Jordan and J. von Neumann, *Ann. Math.* 36, 719 (1935).
- [55] I. Białynicki-Birula, *Phys. Rev. D* 49, 6920 (1994).

- [56] D. A. Meyer, J. Stat. Phys. 85, 551 (1996).
- [57] P. Arrighi, Natural Computing 18, 885 (2019).
- [58] D. Bohm, Phys. Rev. 85, 166 and 180 (1952).
- [59] D. Dürr, S. Goldstein and N. Zanghi, J. Stat. Phys. 67, 843 (1992).
- [60] R. W. Spekkens, Phys. Rev. A 75, 032110 (2007).
- [61] N. Harrigan and R. W. Spekkens, Found. Phys. 40, 125 (2010).
- [62] B. S. Cirel'son, Lett. Math. Phys. 4, 93 (1980).
- [63] S. Popescu and D. Rohrlich, Found. Phys. 24, 379 (1994).
- [64] H. S. Poh, S. K. Joshi, A. Cerè, A. Cabello and C. Kurtsiefer, Phys. Rev. Lett. 115, 180408 (2015).
- [65] B. Hensen et al., Nature 526, 682 (2015).
- [66] P. Grangier, G. Roger and A. Aspect, Europhys. Lett. 1, 173 (1986).
- [67] A. Gonen, Universe24: a discrete simulator of local physical laws (Reality Theory), Zenodo (2026), doi:10.5281/zenodo.22738746 (concept DOI, resolving to the latest version).

## A Auxiliary proofs

**Proof of Theorem 1.** A map of the six Ports that keeps opposite Ports opposite permutes the three axes ( $3!$  choices) and then fixes a sign on each ( $2^3$  choices), and every such choice is realised by one signed permutation matrix; so the group is the hyperoctahedral group  $B_3$ , the symmetry group of the cube and of the octahedron of the Ports, of order 48. The determinant of a signed permutation matrix is  $\pm 1$  and multiplicative, and it is  $-1$  on the reflection in one coordinate plane, so it is onto  $\{+1, -1\}$ ; the kernel of a homomorphism onto a group of two elements has index 2, so there are 24 rotations and 24 reflections. A rotation of the cube permutes its four body diagonals; the only signed permutations fixing all four diagonals as lines are the identity and the central inversion, whose determinant is  $-1$ , so the action of the kernel is faithful, and both groups having order 24, the rotations are the symmetric group on the four diagonals. A pseudoscalar column goes to  $\det(g)$  times itself by definition. That the six operations commute with the 48 up to the declared ties is the statement above, not part of this theorem.

**Proof of Theorem 2.**  $\sum_i (wa_i)^2 / (\mathfrak{m}A) = w^2 \sum_i a_i^2 / (\mathfrak{m}A) = w^2 / \mathfrak{m}$ . The conjugate transpose maps the outputs  $(wa_i, \mathfrak{m}A, p + t_i)$  to  $\sum_i a_i (wa_i)$  at the phase  $p$  on the input direction with multiplicity  $\mathfrak{m}A \cdot A$ , that is  $(Aw, A^2 \mathfrak{m}, p) \sim (w, \mathfrak{m}, p)$ ; for two inputs at one splitter the cross terms carry phases differing by  $N_\phi/2$  (a reflection's quarter turn taken forward and its reverse taken back) and cancel in the normal form, which the check with the pairs (20, 21), (3, 4, 5) and (119, 120, 169) confirms integer by integer [3] (with the turns kept instead of reversed the inputs are not returned), and the engine's merge reproduces on its own rows [2]. For  $U_s$  the diagonal of  $U_s^\top U_s$  is  $C'^2 + S'^2$  and the off-diagonal  $C'S' - S'C' = 0$  in integers.

**Proof of Theorem 3.**  $F$  maps a basis element of  $Q$  to a basis element, the Node moved by an offset that is a function of  $D$  and of the age, the history's; with the history fixed it is injective on the basis.  $S$  absorbs the arriving row and re-emits its weight, multiplicity and phase on its outputs at age 0, an event; on  $Q$  it is injective by Theorem 2 (the conjugate transpose inverts it on the weights, multiplicities and phases); on  $Q$  there are no two rows differing only in age, and their ages are the history's, the count since this event. (The engine's record of the event, a host diagnostic read by no rule and compared with nothing, is the history the statement fixes; it keeps the age the row arrived with for the books and the replay.)  $R$  is injective because  $U_{s,t} = U_s \text{diag}(1, v(t))$ , the turn a bijection and  $U_s^\top U_s = n_s I$ .  $M$  replaces a representative by the normal form of the same element of  $M_r$ , the identity on the module. A composition of injective linear maps is injective. The collision permutes the directions of the amount-1 rows at a Node as a function of the whole set present [2], a bijection of states and not a linear map, which is why it is excluded from the linear statement. The engine's inverse interval is the witness on worlds without a measured event [2]; with splitters the statement is the algebraic one above.

## B Technical details

**The tables and their rounding.** interval  $t \in \mathbb{Z}$ . The circle is  $\mathbb{Z}_{N_\phi}$  with  $4 \mid N_\phi$ . The tables are  $C_{N_\phi}[p] = \text{round}(256 \cos(2\pi p/N_\phi))$  and  $S_{N_\phi}[p] = C_{N_\phi}[p - N_\phi/4]$ , so  $C_{N_\phi}[p + N_\phi/2] = -C_{N_\phi}[p]$  and  $S_{N_\phi}[p + N_\phi/4] = C_{N_\phi}[p]$  exactly; the norm  $C_{N_\phi}^2 + S_{N_\phi}^2$  lies in  $[65536 - 351, 65536 + 361]$  for every power of two  $N_\phi \leq 65536$  (at  $N_\phi = 64$  the range is  $-88$  to  $+237$ ) [5]. The half-angle tables are  $C'_{N_\phi} = C_{2N_\phi}$ ,  $S'_{N_\phi} = S_{2N_\phi}$ . Directions  $D$  are the six headings and declared primitive integer vectors; the flight table gives, per direction  $D$  and age  $\tau$ , the Node offset  $\text{pos}_D(\tau)$  on the digital line of  $D$ , at most one Link per interval, and Euclidean speed  $N_l|D|/T_D$  with  $T_D = \lfloor \sqrt{3}|D|^2 N_l^2 \rfloor$  and the flight scale  $N_l = 64$ : within 0.8 percent of  $1/\sqrt{3}$  for every  $D$ , exact on  $(1, 1, 1)$  [2]. The Mach-Zehnder record's total over the 64 birth phases, computed from the tables and not a reading, takes eight values from  $65448/65536$  to  $65773/65536$  [4]; the normalisation by  $C_K$  in Eq. (9) makes the click's probabilities sum to one at completion, and the paper claims exact unitarity nowhere.

### The symbols.

Table 6: The symbols of this paper, under the one rule: physics' letters keep physics' meanings, the law's grains are  $N$  with a subscript, a world key is in code font.

| Symbol                                         | Kind                                               | Name                                                                                                                                                                                                                                                                                                                               |
|------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $E, E_0, m, \mathbf{p}, c, v, \beta$           | scalars, one vector                                | the energy, the rest energy $E_0 = mc^2$ , the mass $m = N_l N_w M$ in label units, the momentum, the limit speed $c = 1/\sqrt{3}$ Links per interval, the speed, $v/c$                                                                                                                                                            |
| $h, f$                                         | scalars                                            | the quantum of action ( $E = hf$ ), the frequency                                                                                                                                                                                                                                                                                  |
| $M, N_l, N_w, N_\phi, N_D, N_\theta, N_u, N_t$ | scalars                                            | the content (a count of units), the label's scale, the width, the phase circle, the direction bound, the fan's angular grain, the birth wheel, the cosine tables' scale: the grains of the lattice, its definition lines (the law's documents write $Q, S$ and $N$ for $N_l, N_w$ and $N_\phi$ until their writers' next pass [8]) |
| $\Phi; s, r, d, a, e$                          | operator; scalars                                  | the interval's map; one accumulator of the state, its rate, its wall ( $d_p$ the drive's), its cap, its event count                                                                                                                                                                                                                |
| <b>a</b>                                       | vector                                             | the label flow read at a Node                                                                                                                                                                                                                                                                                                      |
| $\mathbf{G}, U_s, R(f), R(o_A, o_B)$           | a matrix, a matrix written plain, a form, a scalar | the click's Gram matrix, the label rotation (Section 6's matrix, written plain there), the click's reading (weight), a cell's weight                                                                                                                                                                                               |
| $D, \mathbf{u}_D, T_D, S_1, K$                 | an integer vector written plain, a vector, scalars | a direction, its unit vector at the scale $N_l$ , its resolution, its Manhattan length, the number of directions of a fan (the ladder's $K$ , the count of a record's cells in $C_K$ and $b_K$ , is Section 6's)                                                                                                                   |
| $u, \tau, b_k, C_k, C_K$                       | scalars                                            | the birth wheel's value, the age, a rung of the ladder, a cumulative weight, the total                                                                                                                                                                                                                                             |
| $a_r, a_\tau, [n, d]$ pairs                    | scalars, pairs                                     | the presence a body read, the age moment it read, the declared rates (suspension, clock, release) as keys                                                                                                                                                                                                                          |
| $\lambda, Q, \theta$                           | scalars, a module                                  | a wavelength, the quotient of the rows' module that forgets the age (Theorem 3), an angle                                                                                                                                                                                                                                          |
| $z, G, \rho$                                   | scalars                                            | the redshift, Newton's constant, a family's charge per unit of content                                                                                                                                                                                                                                                             |
| <b>m</b>                                       | a row's attribute, in code font                    | a row's multiplicity                                                                                                                                                                                                                                                                                                               |

| Symbol                      | Kind                                                        | Name                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $q, f, s, e, r, d, H, S, W$ | letters with a second meaning, named where the text uses it | a source's release $q$ (Sections 4 and 5) beside the deceleration parameter; a record's count vector $f$ in $R(f)$ beside the frequency; the turn's steps $s$ in $E = hs$ beside the accumulator; Euler's $e$ beside the event count; the radius $r$ beside the rate; the pair's denominator, a phase difference in steps and the derivation's density $d$ beside the wall; Shannon's $H$ of the entropy identity; the CHSH sum $S$ , and Boltzmann's $S$ and $W$ |
| $a, \varrho$                | scalars                                                     | the cap of Eq. (1), or, where the text says so, an acceleration or a party's setting; the tables' radius $N_t - \sqrt{2}/2$ (Theorem 6)                                                                                                                                                                                                                                                                                                                           |

## C Reproduction

The code, the worlds of series L with their expectation file written before the runs, the reading tool that replays a run's register into its world list, the check scripts of the exact computations with their outputs, the runs' summary and the two figure scripts are archived [67]; the register entry of each series carries every run's world, duration and source fingerprint, and NUMBERS.md beside the manuscript maps each number of this paper to its source [5]. The interpreter is Python 3.14, no floating point in any physical module, the runs seconds each. The register's runs carry the fingerprint `ff5c382d672f`, the figures' runs `731d0f56c9f9` (`figures/summary.json` [5]), the two cone worlds of `L7 4bf55a62e6fd`. The archived version cited at submission is the tag `paper-2026-09-22` on the merge commit of the pull request that carries this cut, one tree whose history holds the trees named above and the log records this paper cites; it replaces the concept DOI below. Every record, note, design and register row this text cites is on the tree of this cut (main merged into it at `87c7ec6b`).

The long form of this manuscript, with the derivation record of every formula, is the tree's paper at the commit `c15c1174`, the forty-page cut before it at `b0d1ebf7`; this paper is the cut to what is proved, measured after a detector or replicated, built around the general formula.

**The seven confirmations.** The pace: series Q, 290 of 290 primitive directions ( $|a| + |b| + |c| \leq 6$ ) clicking at the derived interval, Node and face, the pace 0.5718 to 0.5893 at finite ages; L7, the cone. The click's power: series L, the (3, 4) split, 63/1 at  $N_\phi = 64$ , 31/1 at 32, 125/3 at 128, the far pair's cells 27, 5, 5, 27. The interference of one record: Mach-Zehnder, 64/0 equal arms, 0/64 half turn, 32/32 quarter turn over 64 births. The exact marginals: the pair at  $N_\phi = 64$ , 32/64 for each party in every bin. GHZ: the four allowed triples 16 each, the products +1 and -1, the others 0. Malus: A12, 128 of 256 at 45 degrees, 0 of 256 crossed, 219 of 256 at 22.5. Light beside a mass: series K, three worlds, the deflection 0.000 pixel, the delay 0.00 interval [4, 9].

**The transition from ordinary physics to modern algebra, dated.** The twenty-four steps of [22], each a thing of physics becoming an object of the law: (1) 2026-09-17, the shared quantum resource deleted, locality without exception; (2) 09-17, the amplitude a phase and a conserved content, Born's rule a declared table; (3) 09-17, a force a catalog entry read at a meeting; (4) 09-18, the Born table computed from  $N_\phi$ , a clock content, a Node its six Ports; (5) 09-19, the law of events, everything from vector operations; (6) 09-19, a quantum an integer row on the record, the collision a permutation, the click the one one-way border, an open face a detector; (7) 09-19, the tables from the keys, a detector's reading the moments of order 0, 1, 2, the push one bilinear form, charge a rational per unit of content; (8) 09-19,  $E = hf$  the cost of a release; (9) 09-20, a force a column with a sign and a lifetime, the coupling a signed inner product; (10) 09-20, two kinds of readings, an emitter in and a detector out; (11) 09-20, the masses and charges the initialisation, the compact quantised, the scale free; (12) 09-20, the amplitude law, a quantum a record, the click chosen by the wheel, the wave the histogram of clicks; (13) 09-20, every count an accumulator on its reader's own record; (14) 09-20 to 21, the Doppler the count of rows a mover crosses; (15) 09-20, the fan a width of the law, the exact phase at the click, the pins detector readings only; (16) 09-21, the whole law one vector operation, the six verbs, the three tests; (17) 09-21, the record an element of  $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ , the weight one bilinear form, Born's rule the unique positive quadratic form, the complex numbers leave; (18) 09-21, the lattice

the translation group, the 48 with the hand,  $c$  the norm of the flight operator; (19) 09-21, the readings by type, a formula gives and a run proves, the dictionary; (20) 09-21, the Lorentz factor a root refused as a seventh verb, the covariant readings kept beside the law; (21) 09-21 to 22, the clock's word the age moment, Newton and Poisson after a detector; (22) 09-22, a detector's clock a member of the age wall at coefficient 1; (23) 09-22, the click theorem, a click the passage of information Node to Node, the click families' transformations the Lorentz group up to scale; (24) 09-22, the framing, Inside and Outside, matches nature and never is nature.

**Use of AI tools.** The code, the design documents, the check scripts and the drafts of this manuscript were produced with AI coding agents under the author's direction and review; the author verified every number against the archived runs and is responsible for the whole text; no AI system is an author. [The tool and version are named at submission.]

## D The families in the algebra, and the check of the roads

Table 7: Every family of the register [10]: its declared integers ( $h$ , the charge  $\rho$ , the columns, the lifetime  $L$ , the circle, the hand; the content  $M$  on the measured worlds), its algebraic object (the frame's section 9 [25]; [22]), when it entered (the record), the verbs that act on it (1 translation, 2 bilinear form, 3 group-ring addition, 4 permutation, 5 evaluation, 6 division) and the click that reads it, by kind. "Not named" is a cell the tree does not fill.

| Family                                                      | The declared integers                                                                                                                                | The algebraic object                                                                                                                                                                  | When it entered                                                                                  | The verbs  | The click that reads it                                                                                   |
|-------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------------------------------------|
| <b>light</b> (the photon)                                   | $h = 1$ ; no charge; the circle turns; $M$ per unit 1; the amplitude worlds' births $2^{20}$ to $2^{33}$                                             | a free row, an element of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ born at the lamp, its amount the coefficient, its direction on the fan under the 48; its phase per Link the declared turn | the amplitude a phase and a content, 2026-09-17 (entry 2); $E = hf$ , 09-19 (entry 8)            | 1, 2, 3, 5 | the two-slit pixels (L2b), the Mach-Zehnder ports 64/0, series Q's 290 of 290 face clicks (DETECTOR)      |
| <b>e</b> (the electron)                                     | $h = 0$ ; $\rho = -15$ ; the circle turns; $M = 1836$ (atoms, bohr), 1 (gallery)                                                                     | the content on $\mathbb{Z}$ ; the charge one entry of the coupling matrix; the momentum an integer 3-vector; the phase on $\mathbb{Z}_{N_\phi}$                                       | the charge a rational per unit, 09-19 (entry 7); the masses the initialisation, 09-20 (entry 11) | 1, 2, 6    | series S's face clicks 369, 345 (DETECTOR); Bohr's ratio (row 6) not run                                  |
| <b>beta, w</b> (the decay's electron and boson)             | $h = 1$ ; a whole charge $-7344$ per unit of amount; $w$ 's lifetime 1, no circle; <b>beta</b> 's circle turns, its hand $-1$ on the world <b>wu</b> | a free row pushing by its label; the lifetime an integer wall; the hand a pseudoscalar in $\mathbb{Z}_2$                                                                              | the catalog, record 30; one definition per family, record 113 (09-20)                            | 1, 2, 6    | the neutron's decay clicks, series J1, 64 clicks, width over median 0.036 (DETECTOR, row 8a)              |
| <b>p</b> (the proton)                                       | $h = 0$ ; $\rho = [1, 1]$ (a winding), 4 on the atoms and binding worlds; no circle; $M = 1834, 1836$                                                | the content on $\mathbb{Z}$ ; the charge a rational pair in the coupling matrix; clicks by no phase                                                                                   | the catalog, record 30; one definition per family, record 113 (09-20)                            | 1, 2, 6    | series N's border clicks: the deuteron's held bond 0.109 percent, the alpha's 2.0 (DETECTOR, rows 7a, 7b) |
| <b>n</b> (the neutron)                                      | $h = 0$ ; no charge; no circle; $M = 1837, 1839$                                                                                                     | the content on $\mathbb{Z}$ ; the momentum an integer 3-vector                                                                                                                        | as above                                                                                         | 1, 2, 6    | series J1's decay clicks (row 8a); series N (row 7a) (DETECTOR)                                           |
| <b>nuclear, glue, bond</b> (the strong column and the bond) | $h = 0, 0, 1$ ; the column <b>strong</b> $\sigma = 10000$ with the sign $-1$ (7000 on one nucleus world); the lifetime 3; no circle                  | the column's signed scalar per unit, one entry of the coupling matrix; the lifetime an integer wall                                                                                   | a force a column with a sign and a lifetime, 09-20 (entry 9)                                     | 2, 6       | series N's border clicks (DETECTOR, rows 7a, 7b)                                                          |
| <b>u</b> (the up quark)                                     | $h = 0$ ; $\rho = 1224$ ; no circle                                                                                                                  | the content on $\mathbb{Z}$ , the charge on $\mathbb{Z}$                                                                                                                              | the catalog, record 30 (09-20)                                                                   | 1, 2, 6    | no detector reading registered (the quarks worlds)                                                        |

| Family                                                                 | The declared integers                                                                                                                                                                        | The object                                                                                                                                                                             | When                                                                                                                       | Verbs   | The click                                                                                                                                    |
|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------|
| nu, (the trino)                                                        | <b>nubar</b> $h = 0$ ; no charge; no content per unit (row 8c); the circle turns; the hand +1 ( <b>nubar</b> ), -1 ( <b>nu</b> ) on the world <b>nu_hand</b> ; $M = 4096$ on the weak worlds | the hand a pseudoscalar in $\mathbb{Z}_2$ under the 48, kept by the rotations and negated by the reflections; the phase on $\mathbb{Z}_{N_\phi}$                                       | the hand, <b>hand-v1</b> [22]                                                                                              | 1, 4, 5 | series J2, 16 of 1024 with 0 behind (DETECTOR, row 8b); the massless input refuted (row 8c)                                                  |
| m, probe, (the massive bodies)                                         | <b>mass</b> , $h = 0$ ; $\rho = 0$ ; no circle; $M$ from 1 to $2^{26}$ as declared per world                                                                                                 | the held content on $\mathbb{Z}$ ; a body's momentum an integer 3-vector; its counts accumulators ( $s, r, d$ ); the age wall's member                                                 | the content, record 15 of 09-19; the masses the initialisation, 09-20 (entry 11); the clock's word, 09-21 to 22 (entry 21) | 1, 2, 6 | D3's births at the line, 138 of 139, $T(24)/T(12) = 1.997$ ; series T's 1.907; series X's $k = 0.9089$ ; series G2's $z = 0.2636$ (DETECTOR) |
| q (the test charge)                                                    | $h = 0$ ; $\rho = [0, 1]$ , $[-1, 2]$ on the coupling worlds; the circle turns; $M = 2^{24}$                                                                                                 | the charge a rational pair in the coupling matrix                                                                                                                                      | the charge a rational per unit, 09-19 (entry 7)                                                                            | 2, 6    | no detector reading registered (the coupling worlds)                                                                                         |
| sa, (the pair's choosers)                                              | <b>sb</b> $h = 0$ ; no charge; the circle turns                                                                                                                                              | a setting's integer matrix $U_s$ in $M_2(\mathbb{Z})$ acting on the arms' label pair; the pair's record in $\mathbb{Z}[\mathbb{Z}_{N_\phi}] \otimes \mathbb{Z}^2 \otimes \mathbb{Z}^2$ | the record an element of $\mathbb{Z}[\mathbb{Z}_{N_\phi}]$ , 09-21 (entry 17)                                              | 2, 3, 5 | the pair's clicks, $S = 2.75$ at $N_\phi = 64$ , the marginals 32/64 (DETECTOR, row 1a)                                                      |
| the apparatus (wall, screen, counter, apparatus, carrier, detector, d) | $h = 1$ ; no charge; the circle turns (d no)                                                                                                                                                 | the click's readers: a detector's set of Nodes, its own count the age wall's member at coefficient 1, its reading the moments of order 0, 1, 2                                         | the readings the moments, 09-19 (entry 7); a detector's clock, 09-22 (entry 22)                                            | 5, 6    | every DETECTOR row: they are the clicks                                                                                                      |
| the sources (s; the 24 sources; the 24 star sources)                   | $h = 0$ or 1; no charge; the circle turns; $M = 64$ , 1024, 4096, 8192                                                                                                                       | a lamp's counts table on $\mathbb{Z}_W$ (the birth wheel); a throw's direction on the fan                                                                                              | the amplitude law, the wheel, 09-20 (entry 12)                                                                             | 1, 6    | series G2's $z = 0.2636$ (row 4b); series T's lamp; the hubble worlds' $q_0 = -0.108$ (row 3) (DETECTOR)                                     |
| mu (the muon, the covariant worlds)                                    | inline: $h = 0$ ; $M = 13248$ (series S)                                                                                                                                                     | the record's $W = E'^2 + 3\mathbf{p} \cdot \mathbf{p}$ with $E'$ the whole root by comparisons (the identity, a hypothesis beside the law)                                             | the covariant readings, 09-21 (entry 20)                                                                                   | 1, 2, 6 | the 64th self-creation at 70 and 124 (GAMEBOARD); its products' face clicks (DETECTOR, series S)                                             |
| <b>matter</b> (the massive rows)                                       | inline: $h = 0$ ; the rows' content per unit                                                                                                                                                 | a paid row, its content per unit on $\mathbb{Z}$ , its momentum content $\times \mathbf{u}_D$                                                                                          | the massive rows, series W (09-21)                                                                                         | 1, 6    | series W's first-click ages 815 and 876 at the pixels (DETECTOR)                                                                             |

Table 8: The check of the roads (record 817): one row per formula the paper arrives at, the inputs its chain used, the file and line where the chain starts, and the word used. No row lists Einstein's or Lorentz's formula among its inputs; “derived” stands in no row. The sources: the click frame on **main** at 70e9781a [25] and Einstein Outside on **main** at e8432e6c [21], whose section III is the same check made in the source.

| The formula arrived at                                      | The inputs its chain used                                                                                                        | Where the chain starts                                                             | The word                                                                                        |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| $W = E_0'^2 + 3\mathbf{p} \cdot \mathbf{p}$ to second order | (A1), (A2); the walk's invariant $\cos \omega = \cos m \cos \kappa$ ; the Planck map on <b>main</b> ; the unit $E' = E/c^2$ [23] | the click frame, section 7 (the invariant :748, the unit :795); [23]               | shown to second order (the conversion's identity); declared for the rows' dynamics (record 270) |
| the boost, Lorentz's form                                   | (A1); two legs at one Node per interval; $k_{AB} = k_{BA}$                                                                       | the click frame, section 2 (b) :278–297; section 0 :84–132 (the group up to scale) | follows; Lorentz's form recovered, the scale $r$ free; (A2) fixes $r$ at $\sqrt{1-v^2}$         |

| The formula                                                                                                                                            | The inputs                                                                                          | Where                                                                                  | The word                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| the place-to-place factor<br>$k_{XY} = (r_Y/r_X)(1 - \mathbf{s} \cdot \mathbf{v}_X)/ (1 - \mathbf{s} \cdot \mathbf{v}_Y)$<br>$1/\gamma$ , the rate $r$ | (A1), (A3), the counts; $r$ a free symbol<br><br>Theorems 1, 2; $r$ read from two factors           | Einstein Outside, Theorem 1 :355<br><br>Einstein Outside, II.1 :647                    | follows (a theorem of A1)<br><br>follows in $r$ ; $\sqrt{1-v^2}$ recovered under the line, shown to second order under (A2), declared under the identity, 1 on the law  |
| the composition $w = (v_R - v)/(1 - vv_R)$<br>$E^2 = E_0^2 + c^2 p^2$                                                                                  | Theorems 1, 2, (A3)<br><br>$E = E_0/r$ , $p = Ev$ , the line of II.1; the frame's section 7 for $W$ | Einstein Outside, II.3 :733<br><br>Einstein Outside, II.6 :887                         | follows, $r$ -free (Einstein's form recovered)<br>follows as the line; $W$ shown to second order; exact by declaration under the identity; $E_0 = mc^2$ a declared unit |
| the Outside step, place apart and count apart of a detector, Einstein's step its most general form                                                     | (A1), (A3), the Inside step, Theorems 1, 2                                                          | Einstein Outside, section 3 (a) :481 (Einstein's form named at :497 as the comparison) | follows; Einstein's form recovered where $r$ is supplied, never an input                                                                                                |
| the quantum of the step above                                                                                                                          | (A1), (A3), the drive's one Link per self-creation, Theorem 2 (a)                                   | Einstein Outside, section 3 (b) :510                                                   | follows (a theorem; no run)                                                                                                                                             |
| Newton's step as the limit                                                                                                                             | Theorem 4, the shell mean, $v \ll c$                                                                | Einstein Outside, section 3 (c) :554                                                   | recovered (the limit)                                                                                                                                                   |